

Probability Without Randomness: Statistical Emergence from Phase Admissibility

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Phase Theory Series — Paper 5 of 8 | Preprint. Under review. January 2026.

Subject areas: Foundations of Physics / Quantum Foundations / Philosophy of Physics

Keywords: phase admissibility, Born rule derivation, emergent probability, determinism, quantum foundations, phase ontology, coarse-graining, ensemble statistics, no-collapse, hidden variables

Abstract

Probability is traditionally assumed to be a fundamental feature of physical law — irreducible, intrinsic, and axiomatic. We demonstrate that within the framework of Phase Theory, probability is not primitive and does not reflect intrinsic randomness. Instead, probabilistic behavior emerges as a statistical description of ensembles of admissible phase configurations under coarse-grained access and environmental coupling. The Born rule and related probabilistic structures are shown to arise as effective limits of phase admissibility, without invoking stochastic axioms, collapse postulates, or epistemic randomness. Phase Theory contains no random variables, no probability measure as a primitive, and no irreducible indeterminism. Probability is a derived quantity — a ratio of admissible phase resolutions over an ensemble — whose geometric origin in configuration space recovers the full predictive apparatus of quantum mechanics. We develop the complete axiomatic basis for this derivation, prove that the Born rule emerges from phase-area weighting without additional postulates, demonstrate that Bell inequality violations are accommodated without nonlocal probability, resolve the tension between determinism and unpredictability, and address the ontological status of probability across classical, quantum, and cosmological regimes. This paper eliminates probability as a foundational concept in physics and replaces it with a structural

account grounded entirely in Phase Theory.

1. The Status of Probability in Physics

Few concepts in the history of science have achieved the foundational status that probability currently enjoys in physics. From classical statistical mechanics through modern quantum field theory, probabilistic statements constitute not merely useful approximations to an otherwise tractable underlying reality but, in the dominant view, irreducible features of nature itself. Understanding how this extraordinary philosophical position was reached — and why it ought to be questioned — is indispensable before any attempt to displace probability from its foundational role can be evaluated on its merits. The historical arc begins with the great synthesis of Boltzmann and Gibbs in the latter half of the nineteenth century, where probability entered physics as an explicitly epistemic device: one assigned probabilities over microstates because one did not know the precise positions and momenta of the $\sim 10^{23}$ molecules composing a macroscopic gas. The probability was in the mind, not in the world. Boltzmann's H-theorem and its descendants were statistical theorems, not dynamical laws, and the underlying Newtonian mechanics remained strictly deterministic. Probability was a concession to ignorance, powerful in its consequences but ontologically modest in its commitments.

The advent of quantum mechanics in the 1920s transformed this situation decisively, and it is arguable that physics has never recovered from the conceptual shock. The Born rule, introduced by Max Born in 1926 in the context of scattering theory, assigns to the squared modulus $|\psi(x)|^2$ of the wave function the status of a probability density for finding a particle at position x upon measurement. Born's original paper treated this as a working

hypothesis, but it rapidly became enshrined as an axiom of the theory. The postulational status of the Born rule is remarkable: it cannot be derived from the other axioms of quantum mechanics (the Hilbert space structure, the Schrödinger equation, the correspondence principle) without additional assumptions, as Gleason's theorem demonstrates by showing that any measure on a Hilbert space satisfying reasonable consistency conditions must take the form $|\psi|^2$, thereby establishing uniqueness rather than derivability. Von Neumann's 1932 axiomatization, followed by Dirac's canonical treatment, placed the Born rule explicitly among the postulates of quantum mechanics, and there it has remained for a century — not derived, not explained, simply given.

Three major interpretive traditions have developed in response to the probabilistic character of quantum mechanics. The frequentist tradition, associated with von Mises and Reichenbach, identifies probabilities with limiting relative frequencies in infinite sequences of identical trials; it provides an operationally clean account but struggles with the idealization of infinite repetition, the reference class problem (which sequence defines the relevant frequency?), and the question of what grounds the assumption that future trials will reproduce past relative frequencies. The Bayesian or epistemic tradition, ranging from de Finetti's radical subjectivism through Jaynes's maximum-entropy methods to the more recent QBist program of Fuchs and Schack, treats probability as a degree of belief — a feature of agents, not of physical systems. This approach dissolves certain paradoxes but raises the question of why the Born rule should constrain rational belief in the specific, rigid way it does, rather than being freely revisable by agents. The propensity interpretation, associated with Popper and later refined by Giere and Humphreys, treats probability as an objective physical disposition of systems to produce outcomes — a kind of irreducible tendency inscribed in nature's furniture. Propensity accounts capture the pre-theoretical intuition that quantum systems "really are" uncertain, but face the objection that

dispositions without categorical bases are explanatorily empty: to say that an electron "has a propensity of one-half to spin up" is to rename the phenomenon without explaining it.

All three traditions share a common feature that Phase Theory calls into question: each treats probability as a primitive, either as an objective feature of the world (propensity, frequentist limit) or as an irreducible feature of rational agents (Bayesianism). In standard quantum mechanics, this primitiveness is further entrenched by the measurement problem — the difficulty of reconciling the deterministic, linear, unitary evolution described by the Schrödinger equation with the apparently discontinuous, nonlinear, irreversible update that occurs upon measurement. The measurement problem is properly understood as a symptom of treating probability as primitive: if the Born rule is simply postulated, then the wave function collapse that implements it is similarly without dynamical explanation, producing a theory that is phenomenologically superb but ontologically incoherent. The Schrödinger equation is manifestly deterministic and linear; the measurement process as described by the collapse postulate is manifestly indeterministic and nonlinear. No resolution of this contradiction is available within the probabilistic framework, because the contradiction is constitutive of it.

The precise question that Phase Theory is designed to answer is therefore not "how should we interpret probability?" but "what physical structure gives rise to probabilistic descriptions at all?" This is a more fundamental question than any of the interpretive debates, because it challenges the premise shared by all interpretations — namely, that probability is the right starting point. Phase Theory begins elsewhere entirely: with the phase field Φ as the sole primitive constituent of reality, and the admissibility functional $I[\Phi] \geq 0$ as the sole governing constraint. From these two elements, without any stochastic axiom, collapse postulate, or probability measure, the full probabilistic

apparatus of quantum mechanics — including the Born rule, interference, contextuality, and Bell correlations — is derived as an emergent, effective description of ensembles of admissible configurations under coarse-grained access. Probability, on this account, is not what the world is made of; it is what an observer with limited access to the world is forced to say about it.

2. Overview of Phase Theory — Axiomatic Foundations

Phase Theory, developed in Papers 1 through 4 of this series, advances a single, radical ontological claim: the physical world is constituted entirely by a phase field Φ , and all other physical structures — particles, forces, spacetime, information — are emergent features of the global behavior of Φ under the constraint of admissibility. The claim is not that Φ is a field in the sense of quantum field theory (an operator-valued distribution on a fixed spacetime background) but that Φ is a more primitive geometric object, a map from a topological substrate M to a target space — typically the group $U(1)$ or a higher-rank gauge group G — from which spacetime itself is emergent. This inversion of the usual explanatory order, treating geometry as prior to the spacetime that supports it, is the signature move of Phase Theory and the source of its unusual power.

The governing constraint of Phase Theory is the admissibility condition: $I[\Phi] \geq 0$, where $I[\Phi]$ is the admissibility functional, a real-valued functional on the space of phase configurations. $I[\Phi]$ is not an energy function whose minimum is sought, nor a Lagrangian whose stationary points are found; it is a constraint that partitions the space of all configurations into the admissible ($I[\Phi] \geq 0$) and the inadmissible ($I[\Phi] < 0$). Physics is the study of admissible configurations and their properties. This is a conceptually clean starting point: there is no dynamics in the conventional sense (no equations of motion derived from an action principle), no Hamiltonian, and no probability. What looks like dynamics is the structured flow of admissible configurations

through the topological substrate under the pressure of global consistency — the enforcement of admissibility across all of M simultaneously. Papers 1 through 3 established that this global consistency enforcement reproduces Maxwell's equations, the Einstein field equations, and the Schrödinger equation as effective descriptions valid in appropriate regimes.

The discrete particle spectrum of the standard model emerges, in Phase Theory, from topological invariants of the phase configuration. Paper 2 demonstrated that the integer-valued topological winding numbers of Φ — the degrees of the map $\Phi : M \rightarrow U(1)$ over closed submanifolds of M — are conserved under admissibility-preserving evolution, and that they are quantized by the topology of the target space G . Charge, spin, and baryon number are not internal symmetry labels but topological winding numbers of different types. The discreteness that quantum mechanics treats as evidence of irreducible randomness — the fact that measurements of spin yield only $\pm\hbar/2$, never intermediate values — is, in Phase Theory, the discreteness of topological invariants: winding numbers are integers, and integers are not the result of random selection from a continuum but of global topological constraints. Paper 3 extended this to the emergence of spacetime metric: the geometry of the effective spacetime experienced by low-energy observers is the geometry of the phase coherence structure of Φ .

The axiomatic base of Phase Theory, as relevant to the present paper, consists of three elements. Axiom PT1 (Phase Ontology): The fundamental constituent of physical reality is the phase field $\Phi : M \rightarrow G$, where M is a compact topological manifold without boundary and G is a compact Lie group. Axiom PT2 (Admissibility): A configuration Φ is physically admissible if and only if $I[\Phi] \geq 0$, where $I[\Phi] = \int_M L(\Phi, \nabla\Phi, \kappa) d\mu$ is the admissibility functional, L is the admissibility density incorporating phase coherence length κ and topological winding constraints, and $d\mu$ is the geometric measure on M . Axiom PT3 (Global Consistency): The physically realized phase

configuration at any time is an element of the admissible configuration space $\text{Conf_adm} = \{\Phi \mid I[\Phi] \geq 0\}$, and its evolution is constrained to remain within Conf_adm . There are no further axioms. In particular, there is no axiom assigning probabilities to configurations, no collapse axiom, and no measurement axiom. These will be derived.

3. Absence of Random Variables in Phase Theory

To establish that Phase Theory contains no random variables, it is necessary first to be precise about what a random variable is. In the standard mathematical formulation, a random variable is a measurable function $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (S, \Sigma)$ from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ — where Ω is the sample space, $\mathcal{F}$ is a σ -algebra of events, and $\mathbb{P}$ is a probability measure satisfying $\mathbb{P}(\Omega) = 1$ — to a measurable space (S, Σ) . The probability space is the primitive structure: without it, a random variable is not defined. The question of whether Phase Theory contains random variables is therefore the question of whether its formalism requires or implicitly assumes the existence of a probability space at the fundamental level.

The Phase Theory configuration space $\text{Conf_adm} = \{\Phi \mid I[\Phi] \geq 0\}$ is a topological space equipped with a geometric measure μ_{adm} derived from the symplectic structure of the phase field, as will be developed in detail in Section 8. This measure space $(\text{Conf_adm}, \mathcal{B}, \mu_{\text{adm}})$, where $\mathcal{B}$ is the Borel σ -algebra induced by the phase topology, is not a probability space, because $\mu_{\text{adm}}(\text{Conf_adm})$ is generically not equal to unity — it is an unnormalized geometric measure (the total symplectic volume of the admissible configuration space). No element of this structure corresponds to the sample space Ω of a probability space, because there is no random trial with an outcome to be selected. The phase configuration Φ is not a value drawn from a distribution; it is the sole existent structure. There is no "randomizer" behind Phase Theory, no hidden probabilistic mechanism that "chooses"

which Φ is realized. The realized Φ is simply the configuration that is globally admissible, given boundary conditions.

The evolution of Φ within Phase Theory is entirely deterministic. Given an initial condition $\Phi_0 \in \text{Conf_adm}$, the global consistency enforcement (Axiom PT3) uniquely determines the sequence of admissible configurations $\{\Phi(t)\}$ for all t , subject to the constraint that each $\Phi(t) \in \text{Conf_adm}$. This is not a stochastic differential equation. There is no Wiener process $W(t)$, no Brownian motion term, no Langevin noise $\eta(t)$ with $\langle \eta(t)\eta(t') \rangle = 2D\delta(t-t')$. The theory that is closest in spirit to incorporating stochastic elements into quantum foundations — Nelson's stochastic mechanics — explicitly introduces a Wiener process as a primitive and derives the Schrödinger equation as a consequence of Brownian motion in configuration space. The Parisi-Wu stochastic quantization program similarly introduces a fictitious stochastic time evolution governed by a Langevin equation, obtaining quantum field theory amplitudes as equilibrium distributions. Phase Theory makes none of these moves: stochasticity is not introduced at any level, not even as a computational tool that is later shown to be eliminable.

The discreteness of physical quantities — the quantization of charge, spin, angular momentum, and particle number — is often cited as the most compelling phenomenological evidence for irreducible quantum randomness: if a measurement of spin must yield one of only two values ($\pm\hbar/2$), then the selection between them must be random. Phase Theory dissolves this reasoning by identifying the source of discreteness as topological, not probabilistic. The topological winding number $n[\Phi] = (1/2\pi) \oint d\Phi$ is an integer-valued invariant of the phase configuration, conserved under continuous deformations of Φ within Conf_adm . A measurement of spin is, in Phase Theory, a resolution of the topological winding sector of the relevant phase configuration. The fact that only integer winding numbers are accessible is not the result of random selection from among integers — it is

the mathematical fact that the degree of a continuous map from a circle to a circle is an integer. Discreteness is topology, not randomness. This distinction from stochastic mechanics and stochastic quantization is fundamental: those programs accept quantization as a fact about probability distributions; Phase Theory explains quantization as a fact about topology.

4. The Admissibility Functional — Formal Structure

The admissibility functional $I[\Phi]$ is the central mathematical object of Phase Theory, playing the role that the Hamiltonian plays in classical mechanics or the action plays in field theory, while differing from both in its logical status. It is not minimized (as in variational principles), not exponentiated (as in path integrals), and not used to generate equations of motion. It is a filter: configurations for which $I[\Phi] \geq 0$ are admitted; those for which $I[\Phi] < 0$ are excluded from the physically realized set. This filtering role makes $I[\Phi]$ a more primitive object than either a Hamiltonian or an action, since the latter two presuppose that all configurations in the configuration space are in principle accessible, differing only in probability weight or energy cost. $I[\Phi]$ determines what exists, not how probable it is for existing things to behave in certain ways.

Formally, let M be a compact, connected, orientable topological manifold of dimension d , without boundary (ensuring global consistency without edge effects), and let G be a compact, connected Lie group with Lie algebra $\mathfrak{g}$. The phase field is a smooth map $\Phi : M \rightarrow G$. For the simplest and physically most important case, $G = U(1)$, and the phase field is a $U(1)$ -valued function on M ; for the full standard model of particle physics, $G = U(1) \times SU(2) \times SU(3)$. Define the admissibility density by:

$$L(\Phi, \nabla\Phi, \kappa) = \kappa^2 |\nabla\Phi|^2 + V(\Phi) - W(n[\Phi]) \quad (4.1)$$

where $|\nabla\Phi|^2$ is the squared covariant gradient of Φ in the metric on M , $V(\Phi)$ is a local potential term encoding self-interaction of the phase field, $W(n[\Phi])$ is a topological weighting term depending on the winding number $n[\Phi]$ of the configuration, and κ is the phase coherence length — the characteristic scale over which phase correlations are maintained. The admissibility functional is then:

$$I[\Phi] = \int_M L(\Phi, \nabla\Phi, \kappa) d\mu_M \quad (4.2)$$

where $d\mu_M$ is the Riemannian volume measure on M . The admissibility condition $I[\Phi] \geq 0$ is the sole selection criterion. The admissible configuration space is $\text{Conf_adm} = \{\Phi \in C^\infty(M, G) \mid I[\Phi] \geq 0\}$, which is a subset of the infinite-dimensional function space $C^\infty(M, G)$. Conf_adm inherits a topology from the C^∞ topology on the ambient space, and it is in this topology that the admissibility measure μ_{adm} will be defined.

The structure of Conf_adm is considerably richer than that of a simple constraint set. Because the winding number $n[\Phi]$ is a topological invariant (integer-valued and continuous in the appropriate sense), Conf_adm decomposes into connected components indexed by winding numbers: $\text{Conf_adm} = \sqcup_{n \in \mathbb{Z}} \text{Conf_adm}^{(n)}$, where $\text{Conf_adm}^{(n)} = \{\Phi \in \text{Conf_adm} \mid n[\Phi] = n\}$. These connected components are the phase sectors of Phase Theory. Phase sectors are not probability amplitudes — they are geometric subsets of the configuration space, distinguished by topological invariants. Within each sector $\text{Conf_adm}^{(n)}$, the phase configuration Φ can vary continuously without changing its sector identity. Transitions between sectors require crossing a topological barrier — a path through Conf_adm that passes through a configuration with $I[\Phi] = 0$ and changes the winding number. Such transitions correspond, in the effective quantum mechanical description, to quantum tunneling events.

The measure $d\mu_M$ on the topological substrate M is the geometric measure inherited from the phase coherence structure of Φ , not the Riemannian measure of any pre-given spacetime. This is an important distinction: the spacetime metric is emergent from Φ (established in Paper 3), so $d\mu_M$ is itself a derived object, determined self-consistently by the global phase configuration. The admissibility condition $I[\Phi] \geq 0$ can therefore be understood as a self-consistency condition: the phase field must cohere with the geometry that it itself generates. This circularity is not vicious but constitutive — it is the Phase Theory analog of the Einstein equations, which self-consistently relate the geometry of spacetime to its matter content. In Phase Theory, the geometry of M and the phase field Φ are mutually constituting, with admissibility as the constraint that enforces their consistency.

A key feature of $I[\Phi]$ that distinguishes it from classical field theory action functionals is the role of the coherence length κ . In standard field theory, there is no such parameter — the field equations are local and do not distinguish short from long-range correlations at the fundamental level (renormalization introduces running couplings, but these are a consequence of integrating out degrees of freedom, not a primitive feature). In Phase Theory, κ is a fundamental parameter encoding the scale over which phase coherence can be maintained against perturbation. When κ is large (long-range coherence), the admissibility condition is tight — many configurations that would be locally admissible fail the global test — and topological constraints are strongly enforced. When κ is small (short-range coherence, appropriate to the classical limit), the admissibility condition becomes approximately local, and the sector structure is effectively invisible to macroscopic observers. The hierarchy of physical phenomena across scales — from quantum coherence at $\sim 10^{-10}$ m to classical behavior at $\sim 10^{-3}$ m and above — is thus encoded in the single parameter κ .

5. Ensemble Formation and the Admissible Ensemble

The concept of an ensemble — a collection of systems prepared in "the same way" — is as old as statistical mechanics, but its precise definition has always been philosophically problematic. In Gibbs's formulation, an ensemble is a hypothetical collection of infinitely many mental copies of a system, one for each possible microstate consistent with the macrostate. The probabilities assigned to ensemble members are either objective frequencies (if the ensemble is real) or degrees of belief (if it is hypothetical). Phase Theory provides a definition of the ensemble that is neither frequentist nor Bayesian but structural: the admissible ensemble is the set of all admissible phase configurations consistent with a given preparation procedure.

Define a macroscopic preparation procedure P as a specification of a finite set of constraints $\{C_\alpha\}$ on the phase configuration Φ , where each constraint C_α corresponds to a macroscopically controllable and measurable condition — setting a magnetic field, preparing a beam with a definite momentum direction, cooling a system to a specified temperature range, and so on. The constraints $\{C_\alpha\}$ are physically implementable and correspond to restrictions on the global phase configuration: each C_α specifies that Φ must belong to a subset $S_\alpha \subset \text{Conf_adm}$. The constrained admissible space is $C_P = \bigcap_\alpha S_\alpha \subset \text{Conf_adm}$, and the admissible ensemble is:

$$E(P) = \{\Phi_i \in \text{Conf_adm} \mid \Phi_i \text{ satisfies all constraints } C_\alpha \text{ of } P\} \quad (5.1)$$

Each element Φ_i of $E(P)$ is a fully determinate, admissible phase configuration — a complete specification of the phase field over all of M . It is not a superposition, not a wave function, not a probability amplitude. It is as definite as a specification of all particle positions and momenta in classical mechanics, and considerably more fundamental, since positions and momenta are themselves emergent from Φ . The admissible ensemble $E(P)$ is

generically infinite-dimensional: it is a subset of the infinite-dimensional function space $C^\infty(M, G)$, and when the constraints C_P underdetermine the full phase configuration (which is the generic case for any physically realizable preparation procedure), $E(P)$ contains uncountably many elements. This is the Phase Theory correlate of the statement that a macrostate corresponds to vastly many microstates.

The non-triviality of $E(P)$ — the fact that it contains more than one element — is not an accident or a limitation of experimental technique. It is a structural feature of Phase Theory. Any physically realizable preparation procedure specifies conditions on the phase configuration in a finite region of M and over a finite time interval. The phase configuration over the remainder of M — the environment, the rest of the universe — is constrained by admissibility but not uniquely fixed by the preparation procedure. This residual freedom is not ignorance about hidden variables; it is the genuine underdetermination of the global phase configuration by any local preparation. Because M is a global manifold, local constraints propagate globally through the admissibility condition, but they do not uniquely determine Φ . The ensemble $E(P)$ is therefore a genuine physical object — the set of all phase configurations that are admissible and consistent with the preparation — not a representation of an agent's ignorance about which configuration obtains.

The relationship between the admissible ensemble and the thermodynamic notion of a macrostate deserves careful attention. In standard statistical mechanics, a macrostate M is defined by specifying a small number of thermodynamic variables (temperature, pressure, volume, magnetization), and the microstate is a point in the classical phase space Γ consistent with M . The Boltzmann entropy $S = k_B \ln W$, where W is the number of microstates consistent with M , quantifies the multiplicity of the macrostate. The fundamental postulate of statistical mechanics — equal a priori probabilities for all microstates consistent with a macrostate — is required to convert W

into a probability distribution. Phase Theory requires no such postulate. The admissible ensemble $E(P)$ is the Phase Theory analog of the Boltzmann macrostate, but the measure on $E(P)$ is not the uniform measure (which is not defined in infinite dimensions without additional structure) but the phase-geometric measure μ_{adm} — the natural measure inherited from the symplectic structure of the phase field. No equal a priori probability assumption, no Boltzmann counting, no ergodic hypothesis is needed. The ensemble structure and its measure are determined entirely by the geometry of Conf_{adm} .

6. Coarse-Graining, Environmental Coupling, and Information Loss

The passage from the admissible ensemble — a set of determinate phase configurations — to a probability distribution over observable outcomes requires an intermediate step: the coarse-graining of Conf_{adm} by the projection onto the space of observable outcomes. This step is not arbitrary or conventional; it is constrained by the physical structure of the measuring apparatus and its coupling to the system. The theory of coarse-graining in Phase Theory is therefore a physical theory, not a mathematical convenience.

Define the space of observable outcomes O as the finite or countable set of distinct, macroscopically distinguishable records that a measuring apparatus A can produce. A measuring apparatus, in Phase Theory, is itself a phase-theoretic system — a configuration $\Phi_A \in \text{Conf}_{\text{adm}}$ with a macroscopic degree of freedom (the "pointer") coupled to the system of interest. The projection $\pi : \text{Conf}_{\text{adm}} \rightarrow O$ is defined by: $\pi(\Phi_i) = O_j$ if and only if the phase configuration Φ_i , when evolved under the admissibility constraint through the measurement interaction with apparatus A , produces apparatus configuration Φ_A with pointer reading O_j . The projection π is many-to-one: many distinct phase configurations $\Phi_i \in E(P)$ produce the same observable

outcome O_j , because the apparatus is sensitive only to certain topological features of Φ_i (those probed by the measurement interaction) and is insensitive to the vast remainder of the phase configuration. This insensitivity is not a design limitation but a structural consequence of the apparatus being a physical system with finite coupling to Φ .

Environmental coupling introduces a further layer of coarse-graining. Define the environment E as the totality of phase degrees of freedom in Conf_{adm} that are not accessible to the apparatus A — the rest of the universe excluding the system S and apparatus A . When S couples to E during the time between preparation and measurement, the admissible ensemble $E(P)$ of S evolves to an ensemble $E(P, t)$ of the composite system $S+E$. The coupling to E is not merely a perturbation; it generically entangles the topological sector of S 's phase configuration with that of E . In Phase Theory, "entanglement" in this context means specifically: the topological sector of Φ_S cannot be specified independently of the sector of Φ_E in the composite admissible ensemble $E(P, t)$. The topology of the composite configuration is a global invariant, not a product of local invariants. This sector entanglement is the Phase Theory account of quantum entanglement, and it explains why environmental coupling degrades coherence: the projection π acts only on Φ_S , so sector cross-terms in the composite ensemble — which contribute to interference when projected onto Φ_S — are washed out when the E degrees of freedom are not resolved.

The information loss associated with coarse-graining is of a specific, non-epistemic character. In classical statistical mechanics, information loss under coarse-graining reflects an agent's decision not to track all degrees of freedom — a conceptually reversible operation in principle, even if practically irreversible. In Phase Theory, information loss under coarse-graining is structural: the coarse-grained observable O_j does not correspond to any subfunctional of $I[\Phi]$ that could, even in principle, be measured by any physical

apparatus. The reason is that the projection π is defined by the apparatus coupling, and any physical apparatus is a phase-theoretic system subject to the same admissibility constraints as the system it measures. A "more refined" apparatus A' that resolves more degrees of freedom of Φ_i will itself have more degrees of freedom coupled to E , producing a larger E that introduces more environmental coarse-graining. This regress has no stopping point: the limit of infinite refinement is not a measurement but the dissolution of the apparatus-system distinction — the limit in which $A' \cong \Phi$ itself, and all distinctions collapse. Information loss is therefore a structural theorem of Phase Theory rather than a practical limitation.

Coarse-Graining Theorem (6.1).

For any preparation P and projection $\pi : \text{Conf}_{\text{adm}} \rightarrow O$, the image $\pi(E(P))$ is a finite or countable set of outcomes $\{O_j\}$, and for each O_j , the pre-image $\pi^{-1}(O_j) \cap E(P)$ is a measurable subset of $E(P)$ with well-defined phase-geometric measure $\mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P)) \geq 0$. The sum $\sum_j \mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P)) = \mu_{\text{adm}}(E(P))$.

Proof. The measurability of $\pi^{-1}(O_j) \cap E(P)$ follows from the measurability of π (which is determined by the apparatus coupling, a physical process preserving the Borel structure of Conf_{adm}) and the measurability of $E(P)$ (which is a closed subset of Conf_{adm} in the phase topology, hence in the Borel σ -algebra $\mathcal{B}$). Non-negativity of μ_{adm} follows from the definition of the geometric measure (Section 8). The partition property $\sqcup_j \pi^{-1}(O_j) = \text{Conf}_{\text{adm}}$, restricted to $E(P)$, gives the normalization by σ -additivity of μ_{adm} . ■

The multiplicity $\mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P))$ is the fundamental combinatorial object from which probability emerges in Phase Theory. It is not a count of discrete microstates (as in Boltzmann) nor a Lebesgue measure on a phase space (as in Liouville); it is the phase-geometric measure on the subset of the admissible ensemble that projects to outcome O_j . The next section

formalizes this into a definition of emergent probability and proves that it satisfies the Kolmogorov axioms.

7. Definition of Emergent Probability — Formal Derivation

We are now in a position to give the formal definition of probability within Phase Theory. The definition is not an axiom — it is a derived definition, justified by the physical account of ensemble formation and coarse-graining developed in the preceding sections. The critical philosophical point is that the word "probability" appears in what follows only as a label for a derived quantity; the quantity itself is defined in terms of measure-theoretic concepts applied to the geometry of Conf_adm , without any invocation of probabilistic primitives.

For a preparation procedure P , a projection $\pi : \text{Conf_adm} \rightarrow O$, and an outcome $O_j \in O$, define the Phase-Theoretic probability of outcome O_j given preparation P as:

$$P(O_j | P) = \mu_{\text{adm}}(\{\Phi \in E(P) \mid \pi(\Phi) = O_j\}) / \mu_{\text{adm}}(E(P)) \quad (7.1)$$

This is a ratio of geometric measures: the measure of the subset of the admissible ensemble that projects to outcome O_j , divided by the total measure of the admissible ensemble. When $E(P)$ is finite, this reduces to the combinatorial ratio $|\{\Phi_i \in E(P) \mid \pi(\Phi_i) = O_j\}| / |E(P)|$, which is the classical definition of probability in equally likely cases. The phase-geometric measure μ_{adm} generalizes this to the infinite-dimensional setting, where counting is replaced by geometric volume in Conf_adm .

Theorem 7.1 (Kolmogorov Axioms from Phase Geometry).

The function $P(\cdot | P)$ defined by equation (7.1) is a probability measure on the outcome space O satisfying the three Kolmogorov axioms: (i) $P(O_j | P) \geq 0$ for all

j; (ii) $P(O_j | P) = \sum_j P(O_j | P) = 1$; (iii) For disjoint outcomes O_j and O_k , $P(O_j \cup O_k | P) = P(O_j | P) + P(O_k | P)$.

Proof. (i) Non-negativity: μ_{adm} is a non-negative measure by definition (it is the Hausdorff measure of appropriate dimension on Conf_{adm} , which is non-negative by construction). $\mu_{\text{adm}}(E(P)) > 0$ by the physical assumption that every preparation procedure specifies a non-empty admissible ensemble. Therefore $P(O_j | P) = \mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P)) / \mu_{\text{adm}}(E(P)) \geq 0$. (ii) Normalization: $\sum_j P(O_j | P) = \sum_j \mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P)) / \mu_{\text{adm}}(E(P)) = \mu_{\text{adm}}(E(P)) / \mu_{\text{adm}}(E(P)) = 1$, by the partition property proved in the Coarse-Graining Theorem (6.1). (iii) Countable additivity: for disjoint outcomes O_j and O_k , the pre-images $\pi^{-1}(O_j)$ and $\pi^{-1}(O_k)$ are disjoint subsets of $E(P)$, and μ_{adm} is countably additive by the measure-theoretic properties of the Hausdorff measure. Therefore $P(O_j \cup O_k | P) = \mu_{\text{adm}}(\pi^{-1}(O_j \cup O_k) \cap E(P)) / \mu_{\text{adm}}(E(P)) = [\mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P)) + \mu_{\text{adm}}(\pi^{-1}(O_k) \cap E(P))] / \mu_{\text{adm}}(E(P)) = P(O_j | P) + P(O_k | P)$. ■

Several features of this theorem demand emphasis. First, the proof invokes no probability axiom — it derives the Kolmogorov structure from the measure-theoretic properties of μ_{adm} and the partition structure of the coarse-graining projection π . Second, the probability measure $P(\cdot | P)$ is conditional on the preparation P ; the conditionality is structural, not epistemic. P is not a degree of belief about which Φ obtains; it is the geometric fraction of the admissible ensemble that projects to each outcome. Third, the derivation is exact, not approximate: the Kolmogorov axioms hold for all P and all π satisfying the physical conditions specified in Sections 5 and 6, not merely in some asymptotic or limiting regime. Probability is genuinely emergent — it emerges fully, not approximately, from the geometry of Conf_{adm} .

The measure μ_{adm} is the natural geometric measure on Conf_{adm} , analogous to but distinct from the Liouville measure on classical phase space. The Liouville measure is defined by the symplectic form on the cotangent

bundle T^*Q of the classical configuration space Q ; it is natural in the sense of being preserved by Hamiltonian flow (Liouville's theorem) and of being the unique (up to scale) measure invariant under the symplectic group action. The measure μ_{adm} is similarly natural: it is defined by the symplectic-like structure on Conf_{adm} derived from the phase field (detailed in Section 8), and it is preserved by the admissibility-preserving evolution of Phase Theory (the Phase Theory analog of Liouville's theorem, proved in Paper 4). The naturalness of μ_{adm} — its derivation from the geometry of Phase Theory rather than its postulation — is the source of the non-primitiveness of probability in Phase Theory. Probability is derived because its measure is derived; its measure is derived because the geometry of Conf_{adm} is derived from the admissibility functional $I[\Phi]$.

The domain of the probability measure $P(\cdot \mid P)$ is the outcome space O , which is the space of macroscopically distinguishable measurement records. This is a finite or countable discrete set (in practice), and it is determined by the physical apparatus — specifically, by the projection π . Different apparatus configurations produce different projections π and therefore different outcome spaces O . The probability measure $P(\cdot \mid P)$ is therefore not a single universal measure but a family of measures parameterized by the pair (P, π) — the preparation and the measurement. This is not a defect of Phase Theory but a feature: the context-dependence of quantum probabilities (contextuality, Kochen-Specker) is explained precisely by this dependence of the probability measure on the projection π , as will be developed in Section 15.

8. Phase-Area Weighting and the Geometric Origin of Measure

The measure μ_{adm} on Conf_{adm} is not postulated; it is derived from the intrinsic geometric structure of the phase field. The fundamental structure in

question is a symplectic-like two-form ω on the configuration space Conf_adm , constructed from the phase field Φ in a manner analogous to the construction of the symplectic form on the cotangent bundle in classical mechanics. This section develops the construction in detail and shows that the weighting of phase sectors by μ_{adm} recovers the $|\psi|^2$ structure of the Born rule.

For the case $G = U(1)$, the phase field $\Phi : M \rightarrow U(1)$ can be locally written as $\Phi = e^{i\varphi}$ where $\varphi : M \rightarrow \mathbb{R}$ is the phase angle. The configuration space Conf_adm is an infinite-dimensional manifold, and its tangent space at a point Φ is the space of perturbations $\delta\Phi$ of the phase configuration. Define the two-form ω on Conf_adm by:

$$\omega(\delta_1\Phi, \delta_2\Phi) = \int_M \text{Im}(\delta_1\Phi^* \cdot \delta_2\Phi) d\mu_M \quad (8.1)$$

where $\delta_1\Phi^*$ denotes the complex conjugate of the perturbation $\delta_1\Phi$, and Im denotes the imaginary part. In local coordinates where $\Phi = e^{i\varphi}$, this becomes $\omega(\delta_1\varphi, \delta_2\varphi) = \int_M (\delta_1\varphi \cdot \partial_t \delta_2\varphi - \delta_2\varphi \cdot \partial_t \delta_1\varphi) d\mu_M$, which is the standard symplectic form on the space of functions $\varphi : M \rightarrow \mathbb{R}$, analogous to the $\sum_i dq^i \wedge dp_i$ of classical mechanics with φ playing the role of q and $\partial_t \varphi$ playing the role of p . The form ω is closed ($d\omega = 0$) by the same algebraic argument as in classical mechanics, and it is non-degenerate on the relevant sector of Conf_adm (away from singular configurations where $\Phi = 0$, which are excluded by the admissibility condition). Therefore $(\text{Conf_adm}, \omega)$ is an infinite-dimensional symplectic manifold.

The phase-geometric measure μ_{adm} is defined as the Liouville measure associated with ω : on a $2n$ -dimensional submanifold of Conf_adm , $\mu_{\text{adm}} = \omega^n / n!$. In infinite dimensions, this is regularized by restricting to finite-dimensional approximations (truncations of the phase field to finitely many Fourier modes) and taking the limit as the number of modes increases, subject to the constraint that the result is well-defined. The admissibility

condition $I[\Phi] \geq 0$ restricts the integration to Conf_adm , and the resulting measure is well-defined and σ -finite under the regularity conditions on $I[\Phi]$ established in Appendix A.

The key computation for the Born rule derivation is the phase-area of a sector. Consider a two-sector system — the Phase Theory analog of a qubit — with Conf_adm decomposing into two sectors $\text{Conf_adm} = S_+ \sqcup S_- \sqcup S_\partial$, where S_+ and S_- correspond to topological winding numbers $+1$ and -1 respectively, and S_∂ is the sector boundary (a set of measure zero). A general admissible ensemble for a preparation P decomposes as $E(P) = E_+ \sqcup E_-$ (up to a set of measure zero), where $E_\pm = E(P) \cap S_\pm$. The phase amplitudes $A_\pm$ are defined by the phase-field decomposition $\Phi = A_+ \Phi_+ + A_- \Phi_-$, where $\Phi_\pm$ are the sector basis configurations. The phase-area of each sector in $E(P)$ is:

$$\mu_{\text{adm}}(E_\pm) = |A_\pm|^2 \cdot \mu_{\text{adm}}(E(P)) \quad (8.2)$$

This equality — the derivation of which is the subject of the next section and of Appendix B — is the geometric origin of the Born rule. It says that the phase-area of the sector corresponding to amplitude $A_\pm$ is $|A_\pm|^2$ times the total ensemble measure, giving a probability $P(\pm | P) = |A_\pm|^2$ by equation (7.1). The $|\cdot|^2$ structure arises not from any arbitrary choice or convention but from the symplectic volume computation: the symplectic form ω is quadratic in $\delta\Phi$ (bilinear in two perturbations), and the volume element $\omega^n/n!$ is therefore a polynomial of degree n in the squared amplitudes $|A_k|^2$. In the lowest-order ($n = 1$, two-sector) case, this polynomial is simply $|A_\pm|^2$, giving the Born rule exactly.

To make the geometric argument concrete, consider the two-sector case with $\Phi = \alpha \Phi_+ + \beta \Phi_-$ where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$ (normalization of the phase amplitude vector). The symplectic form ω , restricted to the two-dimensional complex space spanned by $\{\Phi_+, \Phi_-\}$, is the standard Kähler form on $\mathbb{C}^2$: $\omega = (i/2)(d\alpha \wedge d\bar{\alpha} + d\beta \wedge d\bar{\beta}) = (1/2)(d|\alpha|^2 \wedge d\theta_\alpha + d|\beta|^2 \wedge d\theta_\beta)$ where $\theta_\alpha, \theta_\beta$ are

the phases of α and β . The symplectic volume of the sector $E_+ = \{\Phi \mid |\alpha|^2 \in [r, r+dr]\}$ in the unit sphere $|\alpha|^2 + |\beta|^2 = 1$ is proportional to $|\alpha|^2 dr$, and integrating over all $r \in [0, 1]$ gives the total symplectic volume of E_+ proportional to $\int_0^1 |\alpha|^2 d|\alpha|^2 = |\alpha|^2/2$. Normalizing by the total volume gives $P_+ = |\alpha|^2$ exactly. This computation, generalized to arbitrary numbers of sectors and continuous spectra, is carried out in full in Appendix B.

9. Derivation of the Born Rule from Phase Admissibility

The Born rule, in its quantum mechanical formulation, states that for a quantum system in state $|\psi\rangle = \sum_k A_k |k\rangle$ (expressed in the eigenbasis $\{|k\rangle\}$ of an observable $\hat{O}$), the probability of obtaining outcome k upon measurement of $\hat{O}$ is $P(k) = |A_k|^2$. This rule is a postulate in standard quantum mechanics. We now prove it as a theorem of Phase Theory, using only the admissibility functional, the symplectic measure on Conf_{adm} , and the topological sector decomposition.

Begin with an admissible ensemble $E(P)$ for a preparation P , and let $\{\Phi_k\}$ be a complete set of sector basis configurations — admissible configurations in Conf_{adm} that are pairwise in distinct topological sectors ($n[\Phi_j] \neq n[\Phi_k]$ for $j \neq k$) and that together span the relevant sector of Conf_{adm} (every $\Phi \in E(P)$ can be written as a phase-field superposition $\Phi = \sum_k A_k \Phi_k$ where $A_k \in \mathbb{C}$). Define the sector subsets $E_k = \{\Phi \in E(P) \mid \Phi \text{ lies in topological sector } k\}$. By the Coarse-Graining Theorem (6.1), these subsets are measurable and partition $E(P)$ up to a set of measure zero.

Theorem 9.1 (Born Rule).

For an admissible ensemble $E(P)$ with sector decomposition $E(P) = \sqcup_k E_k$ and phase amplitudes A_k , the phase-geometric measure satisfies $\mu_{\text{adm}}(E_k) = |$

$|A_k|^2 \cdot \mu_{\text{adm}}(E(P))$. Therefore the Phase-Theoretic probability $P(k | P) = |A_k|^2$.

Proof sketch. The full computation is in Appendix B. The key steps are: (1) Restrict the symplectic form ω to the finite-dimensional complex space spanned by the sector amplitudes $\{A_k\}$. This restriction is justified by the sector decomposition: the symplectic form on the full infinite-dimensional Conf_{adm} decomposes, under the sector decomposition, into a sum of contributions from each sector, plus cross-terms localized on the sector boundaries S_{∂} (which have measure zero and do not contribute to the sector volumes). (2) Compute the Liouville volume $\text{Vol}(E_k) = \int_{E_k} \omega^n / n!$ by integrating over the sector E_k in the amplitude space $\{A_k\}$. The integral over the unit sphere $\sum_k |A_k|^2 = 1$ (normalization) with the measure induced by ω gives $\text{Vol}(E_k) = |A_k|^2 \cdot \text{Vol}(E(P))$, using the standard computation of symplectic volumes on complex projective space (analogous to the Fubini–Study metric). (3) Normalize: $P(k | P) = \text{Vol}(E_k) / \text{Vol}(E(P)) = |A_k|^2$. ■

The treatment of quantum interference is an essential component of the Born rule derivation. Standard quantum mechanics treats interference as a consequence of the linearity of the Schrödinger equation: the wave function for a system that "went through both paths" is a superposition, and its Born rule probability is $|A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2 \text{Re}(A_1^* A_2)$, where the cross-term $2 \text{Re}(A_1^* A_2)$ is the interference term. Phase Theory gives a geometric account of this term. When sectors E_j and E_k share a common boundary in Conf_{adm} — a sub-manifold of positive measure where both sector labels are valid (sector overlap, developed further in Section 10) — the partition of $E(P)$ into sectors is not clean: the projection π cannot distinguish E_j from E_k contributions in the overlap region. The symplectic volume computation then includes contributions from the overlap region that correspond precisely to the cross-terms $2 \text{Re}(A_j^* A_k)$ of the standard interference formula. Constructive interference ($A_j^* A_k > 0$ in the relevant direction) corresponds to aligned sector boundaries adding their symplectic volume contributions; destructive interference ($A_j^* A_k < 0$) corresponds to anti-aligned boundaries subtracting. Interference is geometry, not mystery.

The exactness of the Born rule derivation deserves emphasis. Various derivations of the Born rule have been proposed within quantum mechanics itself — Gleason's theorem (1957), which shows that any consistent measure on Hilbert space must take the $|\psi|^2$ form; Deutsch's (1999) and Wallace's (2012) decision-theoretic derivations; and Zurek's (2003) envariance argument. All of these derivations operate within the quantum mechanical framework, presupposing the Hilbert space structure, the unitary dynamics, or the rationality axioms of decision theory. They show that the Born rule is the unique measure consistent with certain assumed properties of quantum mechanics — but they cannot derive it from non-quantum axioms. Phase Theory's derivation is categorically different: it proceeds from axioms that do not presuppose quantum mechanics (the admissibility functional and the topological substrate) and obtains the Born rule as a consequence. The Born rule is not the unique consistent measure on a Hilbert space; it is the unique probability measure derived from the symplectic geometry of the Phase Theory admissible configuration space.

The generalization to continuous spectra replaces the sum $\sum_k$ with an integral $\int dk$, and the discrete sector decomposition with a continuous family of sectors parameterized by k . The phase amplitude A_k becomes an amplitude density $A(k)$, and the Born rule becomes $P(k \in [k_0, k_1] | P) = \int_{k_0}^{k_1} |A(k)|^2 dk$, which is the standard Born rule for continuous observables. The derivation carries through with the substitution of Lebesgue integration for summation in the symplectic volume computation. The full derivation for continuous spectra is given in Appendix B, Section B.4.

It is worth pausing to note the structure of the derivation as a whole. The Born rule has been derived from three ingredients: (a) the admissibility functional $I[\Phi] \geq 0$, which determines Conf_{adm} ; (b) the natural symplectic measure ω on Conf_{adm} , which is determined by the phase field structure; and (c) the topological sector decomposition of Conf_{adm} , which is determined by the

winding number structure of Φ . All three ingredients are consequences of the axioms of Phase Theory (Axioms PT1–PT3); none is imported from quantum mechanics. The Born rule is, in this precise sense, not a postulate of quantum mechanics that happens to be empirically correct, but a theorem of Phase Theory that quantum mechanics approximates.

10. Interference, Superposition, and Sector Overlap

Standard quantum mechanics treats superposition as ontologically fundamental: a quantum system can genuinely be in a superposition of two or more distinct states simultaneously, and this is not a description of our ignorance about which state it is in. Interference is then a consequence of the superposition principle combined with the linearity of the Schrödinger equation. Phase Theory inverts this explanatory order: sector overlap in Conf_adm — a geometric property of the admissible configuration space — is the primitive phenomenon, and superposition and interference are effective coarse-grained descriptions of it. This inversion is not merely a change of language; it produces different predictions (Section 25) and a different account of what is physically happening.

Define sector overlap precisely: two phase sectors E_j and E_k (corresponding to topological winding numbers n_j and n_k respectively) overlap if their boundaries ∂E_j and ∂E_k in Conf_adm share a common submanifold $B_{\{jk\}} = \partial E_j \cap \partial E_k$ of positive phase-geometric measure ($\mu_{\text{adm}}(B_{\{jk\}}) > 0$). Sector overlap can occur even when $n_j \neq n_k$, because the boundary of a topological sector is not a sharp topological partition but a region of Conf_adm where the phase configuration is on the verge of changing its winding number — where the admissibility condition $I[\Phi] \geq 0$ is satisfied marginally ($I[\Phi] \approx 0$). In this boundary region, the coarse-graining projection π cannot cleanly assign the configuration to either E_j or E_k ; the

contribution of the boundary region to both sectors is non-zero, producing the cross-terms in the probability distribution.

Constructive and destructive interference correspond to the relative orientation of the phase fields at the sector boundary. On $B_{\{jk\}}$, the phase fields Φ_j and Φ_k (the representative configurations of the respective sectors) have a well-defined relative phase $\theta_{\{jk\}} = \arg(A_j^* A_k)$. When $\theta_{\{jk\}} = 0$ (aligned phases), the boundary contributions from E_j and E_k add constructively, increasing the symplectic volume of the overlap region and thus increasing the probability of outcomes that depend on both sectors. When $\theta_{\{jk\}} = \pi$ (anti-aligned phases), the boundary contributions cancel (destructive interference), reducing the probability of such outcomes to zero. The intermediate cases produce partial interference. All of this follows from the symplectic volume computation without any appeal to wave function linearity.

The double-slit experiment in Phase Theory provides a canonical illustration. Consider an electron propagating from source S to a detection screen, with two slits A and B in an intermediate barrier. In Phase Theory, the two paths correspond to two sectors E_A and E_B of the admissible configuration space Conf_{adm} for the electron's phase field: E_A consists of admissible phase configurations with winding concentrated near slit A , and E_B near slit B . The detection screen is the projection space O , and the outcomes are positions x on the screen. The projection π maps each admissible configuration to the point x on the screen at which the phase field has sufficient intensity to produce a detection event. The sectors E_A and E_B share a boundary $B_{\{AB\}}$ in Conf_{adm} whose geometry — the relative phase $\theta_{\{AB\}}(x)$ as a function of screen position x — determines the interference pattern. The bright fringes (constructive interference) occur where $\theta_{\{AB\}}(x) \approx 0$; the dark fringes (destructive interference) where $\theta_{\{AB\}}(x) \approx \pi$. The interference pattern is not caused by the electron "going through both slits

simultaneously" — each admissible configuration $\Phi_i \in E(P)$ passes through one definite slit. The pattern arises from the geometric distribution of sector boundaries in Conf_{adm} , which reflects the global topology of the two-slit setup.

11. Unpredictability Without Ontological Randomness

One of the deepest tensions in the foundations of physics is the apparent conflict between determinism — the thesis that the future state of the universe is uniquely determined by its present state together with the laws of physics — and the brute phenomenological fact of unpredictability. Quantum mechanics has been read, by many physicists and philosophers since Bohr, as resolving this tension in favor of ontological randomness: the universe is not deterministic, measurement outcomes are genuinely irreducible, and unpredictability is not a limitation of knowledge but a feature of reality. Phase Theory resolves the tension in the opposite direction, restoring determinism while explaining unpredictability structurally.

Define ontological randomness for a system S : S is ontologically random if its future state is not determined by any complete specification of its present state together with any physical law. The requirement is that no such determination exists — not merely that no agent can compute it, but that it does not obtain in nature. This is the sense in which quantum mechanics, on the Copenhagen and related interpretations, is held to be random: the outcome of a spin measurement is not determined by any specification of the electron's prior state, however complete. Define effective unpredictability for a system S with respect to an agent A : S is effectively unpredictable by A if A cannot, using any practically or even theoretically accessible information about S 's current state, determine S 's future behavior beyond the ensemble statistics.

Phase Theory is ontologically deterministic. Given the full phase configuration $\Phi_0 \in \text{Conf_adm}$ at time t_0 , the global consistency enforcement (Axiom PT3) uniquely determines the phase configuration $\Phi(t) \in \text{Conf_adm}$ for all $t > t_0$. There is no branching, no random selection, no stochastic element. The trajectory $\{\Phi(t)\}$ is a deterministic curve in Conf_adm , parameterized by time and lying entirely within Conf_adm by the global consistency enforcement. This is stronger than classical determinism (Laplacian or Hamiltonian), because the constraint is global: admissibility must hold on all of M simultaneously, so the trajectory in configuration space is not a local flow but a globally consistent evolution.

Theorem 11.1 (Structural Unpredictability).

For any physical agent A modeled as a phase-theoretic system ($\Phi_A \in \text{Conf_adm}$), the measurement interaction between A and a system S constitutes an admissibility constraint on the composite admissible ensemble $E(A+S)$ that prevents A from accessing the full configuration Φ . Therefore, A cannot predict the outcomes of measurements on S beyond the ensemble probabilities $P(O_j | P)$.

Proof. Suppose, for contradiction, that agent A can access the full phase configuration Φ of system S . This access means that A 's phase configuration Φ_A becomes correlated with all topological sectors of Φ_S — i.e., the composite configuration $\Phi_{\{A+S\}}$ resolves the sector boundary structure of Φ_S . But any physical measurement interaction is an admissibility-constrained coupling between Φ_A and Φ_S . This coupling is a constraint on the composite admissible ensemble $E(A+S)$: the set of admissible composite configurations compatible with A having full knowledge of Φ_S would require A to correlate with degrees of freedom of Φ_S that are themselves coupled to the environment E . But the environment E is part of the global phase configuration $\Phi_{\{\text{total}\}}$, and coupling A to E through S means that A must also correlate with E 's sector structure. By induction, complete knowledge of Φ_S requires knowledge of the full $\Phi_{\{\text{total}\}}$, which would require A to equal the

entire universe — a contradiction (since A is a proper subsystem). Therefore, no physical agent can access the full configuration Φ , and prediction is limited to ensemble statistics. ■

This result is fundamentally different from the epistemic limitations of classical statistical mechanics. In classical mechanics, a Laplacian demon — an idealized entity with unlimited computing power and access to all positions and momenta — could in principle predict all future states with certainty. The unpredictability of classical statistical mechanics is purely a consequence of computational and observational limitations that could, in principle, be overcome. In Phase Theory, the impossibility of complete access to Φ is not a contingent limitation but a structural consequence of the admissibility constraint: any system that measured Φ completely would need to be as large as the universe itself, which is physically inadmissible (there is no "outside" from which to measure the global phase configuration, because M is compact and without boundary). Unpredictability is an admissibility theorem, not an epistemic fact.

The implication for the philosophy of science is significant. The limits of predictability in physics are not contingent on our computational resources, our experimental sensitivity, or the development of more powerful theories. They are structural: they are built into the topology of the admissible configuration space by the global consistency enforcement. A more powerful theory will not eliminate the unpredictability, because the unpredictability is a consequence of the most fundamental structure — the global admissibility constraint — not a relic of our current ignorance. This is a new kind of principled limit on scientific knowledge, analogous to but distinct from the Heisenberg uncertainty principle: it applies to all agents equally, regardless of their resources, and it follows from admissibility rather than from the commutation relations of operators.

12. Comparison with Classical Probability Theory

Kolmogorov's axiomatization of probability (1933) is one of the most successful applications of abstract mathematics to empirical science. The Kolmogorov framework — a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with sample space Ω , σ -algebra $\mathcal{F}$, and measure $\mathbb{P}$ satisfying $\mathbb{P}(\Omega) = 1$ — is sufficiently general to encompass virtually all probabilistic reasoning in physics, statistics, and probability theory, and it has the enormous virtue of separating the mathematical structure of probability from its interpretation. The structure is clear; the interpretation remains contested. Phase Theory's contribution is to provide a physical derivation of the Kolmogorov structure, resolving the interpretive ambiguity by grounding probability in geometry.

In the Kolmogorov framework, the probability measure $\mathbb{P}$ is a primitive — it is simply posited as a function satisfying the axioms, with no further explanation of why it takes the values it does. The sample space Ω is also a primitive: it is the set of all possible elementary outcomes, and the identification of what counts as an elementary outcome is not given by the mathematics but by the modeling choices of the statistician or physicist. In quantum mechanics, the probability space is typically constructed from the Hilbert space: Ω is the spectrum of an observable (e.g., the set of eigenvalues $\{\lambda_k\}$ of a Hermitian operator $\hat{O}$), $\mathcal{F}$ is the power set of Ω (or a sub- σ -algebra thereof), and $\mathbb{P}$ is given by the Born rule $\mathbb{P}(\{\lambda_k\}) = |\langle k|\psi\rangle|^2$. The Born rule is the explicit form of $\mathbb{P}$; it is postulated, not derived.

In Phase Theory, the analogous structure is completely derived. The sample space Ω corresponds to the admissible ensemble $E(P)$ — not to the set of outcomes, but to the set of all determinate physical configurations consistent with the preparation. The σ -algebra $\mathcal{F}$ corresponds to the Borel σ -algebra on $E(P)$ induced by the phase topology. The probability measure $\mathbb{P}$ corresponds to the normalized phase-geometric measure $\mu_{\text{adm}}(\cdot)/\mu_{\text{adm}}(E(P))$. None of these is postulated; all are derived from the geometry of Conf_{adm} under the

constraints of Phase Theory. The outcome space O (the set of possible measurement results) is the image of $E(P)$ under the projection π , and the probability of each outcome O_j is the measure of $\pi^{-1}(O_j)$ in the derived probability space. The critical difference from Kolmogorov is that μ_{adm} is not postulated but derived from the symplectic structure of the phase field.

The relationship between Phase Theory and classical ergodic theory deserves special attention. Classical statistical mechanics uses the ergodic hypothesis — the assumption that the time-average of an observable equals its ensemble average — to justify the use of ensemble probabilities for single-system predictions. The ergodic hypothesis is not a theorem of classical mechanics; it is an additional assumption whose truth is system-dependent and difficult to verify. Phase Theory replaces the ergodic hypothesis with a structural result. The admissibility-preserving evolution of a phase configuration traces a path in Conf_{adm} that is dense in the relevant topological sector of Conf_{adm} — a consequence of the global consistency enforcement, which prevents the trajectory from settling into a lower-dimensional attractor (admissibility is a global constraint, not a local equilibrium condition). This phase ergodicity theorem — proved in Paper 4 — implies that time averages and ensemble averages coincide for admissible configurations, without any additional hypothesis. The ergodic hypothesis is a theorem in Phase Theory, not an assumption.

13. Comparison with Frequentist Interpretations of Probability

The frequentist interpretation, developed systematically by Richard von Mises in the 1920s and elaborated by Reichenbach, identifies the probability of an event type as the limit of the relative frequency of that event type in an infinite sequence of independent, identically distributed trials. This definition has the virtue of making probability an objective, empirically accessible

quantity — in principle, one can estimate probabilities by counting — but it faces a cluster of foundational difficulties that have never been satisfactorily resolved within the frequentist framework itself. Phase Theory provides a resolution of these difficulties by grounding probability in the geometry of the admissible configuration space, rather than in the statistics of infinite sequences.

The first difficulty for frequentism is the reference class problem: to what infinite sequence of trials should the probability of a given event be referenced? The same event can be described as an instance of many different classes, and the limiting relative frequency may differ across these classes. A coin flip can be described as a fair coin flip (reference class: all fair coin flips, frequency $1/2$), a flip of this particular coin (reference class: all flips of this coin, frequency unknown), or a measurement of a specific physical system (reference class: all measurements of systems in this macroscopic preparation, frequency determined by physics). The frequentist has no principled way to select the reference class without invoking probability in the selection criteria, producing a circularity. Phase Theory dissolves this problem: the reference class is the admissible ensemble $E(P)$, which is uniquely determined by the physical preparation procedure P . There is no ambiguity about which reference class is relevant — it is determined by the physics.

The second difficulty for frequentism is the idealization of infinite trials. No actual experiment runs infinitely many times; all physical probability statements are made on the basis of finite data. The frequentist deals with this through the law of large numbers, which says that relative frequencies converge to probabilities as the number of trials increases. But the law of large numbers is itself a probabilistic theorem — it says that the probability of the frequency deviating from the probability by more than ε goes to zero as $N \rightarrow \infty$. This is not a difficulty for Phase Theory because the law of large

numbers, in Phase Theory, is a theorem about the geometry of Conf_{adm} . As N identically prepared systems are measured, the relative frequency of outcome O_j is a sample average over N draws from the admissible ensemble $E(P)$. The sample average converges to the phase-geometric mean $P(O_j | P) = \mu_{\text{adm}}(\pi^{-1}(O_j) \cap E(P)) / \mu_{\text{adm}}(E(P))$ with probability one (in the sense of Theorem 7.1), by the geometric ergodicity theorem mentioned in Section 12. The convergence of relative frequencies is a structural result about the geometry of the admissible ensemble, not a postulate about infinite sequences.

The third difficulty for frequentism is its inability to assign probabilities to single events. If probability is defined as a limiting relative frequency, it applies only to repeatable event types — but many physical situations of interest are not repeatable in any meaningful sense. The decay of a specific radioactive nucleus, the outcome of a specific quantum measurement performed once, or the formation of a specific structure in the early universe cannot be embedded in a natural infinite reference sequence without significant idealizing assumptions. Phase Theory handles single events naturally: the probability $P(O_j | P)$ is defined for any preparation P and projection π , regardless of whether the experiment is repeated. It is the geometric fraction of the admissible ensemble that projects to O_j , and this fraction is well-defined for single preparations. Repetition is not required; the ensemble is a theoretical object (the set of all admissible configurations consistent with P), not a collection of actual repeated trials.

14. Comparison with Bayesian and Epistemic Interpretations

Bayesian and epistemic interpretations of probability treat it as a degree of rational belief, updated by Bayes's theorem as new evidence arrives. This approach has great practical power: Bayesian inference provides a coherent framework for learning from data, combining prior knowledge with

experimental evidence, and making decisions under uncertainty. In the quantum foundations context, QBism (Quantum Bayesianism, or more recently Quantum Bettabilitarianism), associated with Fuchs, Schack, and their collaborators, holds that quantum states are not objective features of physical systems but expressions of an agent's beliefs about the outcomes of future interactions with those systems. The Born rule, in QBism, is a normative constraint on rational agents — a rule for belief update — rather than a law of nature.

Phase Theory agrees with the Bayesian tradition on one crucial point: quantum states (in the effective quantum mechanical description) encode an information state — specifically, the phase-geometric measure on the admissible ensemble conditioned on a preparation. But Phase Theory disagrees that this is the end of the story. The Bayesian must answer the question: why is the Born rule the correct normative constraint on rational belief? In QBism, this is answered by appeal to consistency (Dutch book arguments, coherence constraints), but these arguments establish only that a rational agent should have some consistent probability function, not that it should specifically obey the Born rule. The Born rule's specific form — the $|\cdot|^2$ rule rather than, say, a $|\cdot|^1$ or $|\cdot|^3$ rule — requires additional justification. Phase Theory provides this justification: the Born rule is the specific probability measure that arises from the symplectic geometry of the admissible configuration space, and agents should adopt it because it reflects the objective structure of reality, not because of abstract rationality constraints.

The inter-subjective convergence of quantum probabilities — the fact that all rational agents with the same experimental information should agree on quantum probabilities, and do — is naturally explained in Phase Theory but is problematic for pure Bayesianism. If probabilities are purely subjective degrees of belief, agents are free to have different priors and may not

converge even in the limit of infinite data (unless their priors satisfy Mercer's condition or some equivalent regularity assumption). In practice, quantum probabilities converge universally: every physicist using the Born rule gets the same numbers, regardless of their personal priors. Phase Theory explains this convergence as the consequence of an objective physical fact — the phase-geometric measure μ_{adm} is determined by the geometry of Conf_{adm} , which is a feature of the physical world, not of agents' beliefs. The convergence of quantum probabilities is the convergence of rational agents on the objective geometry of the admissible configuration space.

The special constraining power of quantum probability — the fact that it violates certain classical probability identities, such as the classical additivity rule for mutually exclusive but not exhaustive events (Feynman's discussion of quantum probability rules) — is, in Phase Theory, a consequence of the non-classical geometry of the phase configuration space. The symplectic volume measure is not additive over classical partitions: when sectors E_j and E_k have overlapping boundaries $B_{\{jk\}}$ of positive measure, the volume of their union is not the sum of their individual volumes (there are cross-terms from the boundary region). This non-additivity of the geometric measure corresponds exactly to the violation of classical probability additivity in quantum interference experiments. The Bayesian must posit this non-additivity as a special feature of the "quantum" probability norm; Phase Theory derives it from the topology of Conf_{adm} .

15. Quantum Contextuality and Phase-Sector Dependence

The Kochen-Specker theorem (1967) establishes that, in a Hilbert space of dimension ≥ 3 , it is impossible to assign pre-existing definite values to all observables in a manner that is consistent with the algebraic relations between them and independent of the measurement context. This result is often interpreted as ruling out all "non-contextual hidden variable" theories

— theories in which measurement outcomes are pre-determined and independent of what other observables are simultaneously measured. Contextuality is frequently cited, alongside Bell non-locality, as evidence for the irreducibly probabilistic character of quantum mechanics. Phase Theory provides a structural explanation of contextuality that dissolves its role as evidence for randomness.

In Phase Theory, the projection $\pi : \text{Conf_adm} \rightarrow O$ is not a universal, context-independent map; it is defined by the specific measurement apparatus A and its coupling to the system. Different measurement contexts — corresponding to measuring different observables $O_1, O_2, \dots$ — correspond to different projections $\pi_1, \pi_2, \dots$ with different outcome spaces $O_1, O_2, \dots$. These projections induce different partitions of the admissible ensemble $E(P)$: π_1 induces the partition $\{\pi_1^{-1}(O_{1,j})\}$ and π_2 induces $\{\pi_2^{-1}(O_{2,k})\}$. For non-commuting observables (in the effective quantum mechanical description), these partitions are incompatible: the sets $\{\pi_1^{-1}(O_{1,j})\}$ and $\{\pi_2^{-1}(O_{2,k})\}$ do not refine each other, and there is no common refinement that could simultaneously define values for both O_1 and O_2 .

Theorem 15.1 (Contextuality).

For any preparation P and any two non-commuting observables O_1, O_2 (in the effective quantum mechanical description), the corresponding projections $\pi_1, \pi_2 : \text{Conf_adm} \rightarrow O_1, O_2$ do not commute as operations on Conf_adm : $\pi_1 \circ \pi_2 \neq \pi_2 \circ \pi_1$ in general. Therefore, the joint probability distribution $P(O_{1,j}, O_{2,k} | P)$ is not well-defined simultaneously for all j, k ; it depends on the order of measurement.

Proof. The projections π_1, π_2 are defined by the apparatus couplings. Each measurement interaction is an admissibility-constrained coupling that modifies the admissible ensemble: after applying π_1 (measuring O_1), the admissible ensemble is conditioned on the O_1 outcome, producing a reduced ensemble $E(P | O_{1,j}) =$

$\{\Phi \in E(P) \mid \pi_1(\Phi) = O_{\{1,j\}}\}$. The subsequent application of π_2 to this reduced ensemble gives the conditional probability $P(O_{\{2,k\}} \mid P, O_{\{1,j\}}) = \mu_{\text{adm}}(\pi_2^{-1}(O_{\{2,k\}}) \cap E(P \mid O_{\{1,j\}})) / \mu_{\text{adm}}(E(P \mid O_{\{1,j\}}))$. If instead O_2 is measured first, one obtains $P(O_{\{1,j\}} \mid P, O_{\{2,k\}})$ for the conditional probability of O_1 given O_2 . Since π_1 and π_2 induce different (incompatible) partitions of $E(P)$, the reduced ensembles after the first measurement differ depending on which observable is measured first, and so the conditional probabilities for the second measurement differ. This is the mathematical content of non-commutativity, and it is a consequence of the apparatus-coupling definition of π , not an assumption. ■

The Kochen–Specker theorem in Phase Theory becomes a theorem about the incompatibility of projections on Conf_{adm} , rather than a theorem about the impossibility of value assignments. The question "does the system have a definite value of observable O before measurement?" is reformulated as "does the admissible ensemble $E(P)$ lie entirely within a single sector of the partition induced by π_O ?" The answer is generically no — $E(P)$ intersects multiple sectors of π_O for most O — not because the system lacks definite properties, but because the preparation P does not specify sufficient phase information to confine $E(P)$ to a single sector of every possible measurement projection simultaneously. This is not a mystery of quantum indeterminacy; it is a structural consequence of the underdetermination of Φ by any physically realizable preparation, combined with the incompatibility of different measurement projections.

16. Bell Inequalities and Phase-Theoretic Nonlocality

Bell's theorem (1964) is perhaps the most profound and surprising result in the foundations of physics. It proves that no physical theory of the form defined by a triple $(\lambda, \rho(\lambda), \{A(a, \lambda), B(b, \lambda)\})$ — where λ is a hidden variable drawn from a probability space with measure ρ , and $A(a, \lambda), B(b, \lambda)$ are the (deterministic or probabilistic) outcomes of measurements at stations with settings a and b respectively — can reproduce all the predictions of quantum

mechanics. Specifically, any such theory must satisfy Bell inequalities of the form $|\langle AB \rangle - \langle AB' \rangle| + |\langle A'B \rangle + \langle A'B' \rangle| \leq 2$, which quantum mechanics violates (up to the Tsirelson bound of $2\sqrt{2}$). The frequent conclusion is that quantum mechanics must be either nonlocal or irreducibly probabilistic (or both). Phase Theory requires neither.

The critical observation is that Bell's theorem applies to theories of a specific mathematical type: theories with a probability space $(\Lambda, \mathcal{A}, \rho)$ of hidden variables, from which individual instances λ are drawn and which determine (possibly stochastically) the measurement outcomes. Phase Theory has no such structure. The admissible configuration space Conf_{adm} is not a probability space from which instances are drawn; there are no "hidden variables" in the Bell sense; and there is no probability measure $\rho(\lambda)$ on a hidden-variable space. Therefore, Bell's theorem simply does not apply to Phase Theory as a mathematical statement. Bell proved that no theory of type $(\lambda, \rho(\lambda), A, B)$ can reproduce quantum mechanics; Phase Theory is not of this type.

This raises the natural question: how does Phase Theory account for the experimentally observed Bell inequality violations? The answer lies in the global topology of the phase configuration. For a composite system $S = S_1 + S_2$ in an admissible ensemble corresponding to a singlet-like preparation, the phase configuration $\Phi_{\{S_1 S_2\}}$ has a global topological property — specifically, its total winding number is zero ($n[\Phi_{\{S_1 S_2\}}] = 0$). This global constraint means that the topological sector of $\Phi_{\{S_1\}}$ and that of $\Phi_{\{S_2\}}$ are globally correlated: knowing the sector of $\Phi_{\{S_1\}}$ determines the sector of $\Phi_{\{S_2\}}$. This is not because a signal passes between S_1 and S_2 ; it is because the winding number is a global invariant of the composite configuration, defined over the entire manifold M . The correlation is topological, not dynamical.

The joint probability $P(a, b | P)$ for outcomes at the two measurement stations, computed using the phase-geometric measure of the composite admissible ensemble, reproduces the quantum mechanical prediction. For a singlet-like admissible ensemble with measurement settings a, b (unit vectors defining the measurement orientations at the two stations), the joint probability is:

$$P(+, + | a, b) = P(-, - | a, b) = (1 - \cos \theta_{ab})/4 \quad (16.1)$$

$$P(+, - | a, b) = P(-, + | a, b) = (1 + \cos \theta_{ab})/4 \quad (16.2)$$

where θ_{ab} is the angle between a and b . These probabilities give the correlation function $\langle A(a)B(b) \rangle = -\cos \theta_{ab}$, which is the standard quantum mechanical result and violates the Bell inequality. The derivation uses only the phase-geometric measure on the composite admissible ensemble and the global topological constraint $n[\Phi_{S_1 S_2}] = 0$ — no nonlocal signal, no collapse, no hidden variable distribution. Bell's theorem eliminates local hidden-variable theories; it does not eliminate Phase Theory.

The conceptual conclusion is important to state precisely. Phase Theory is local in the sense that no physical influence propagates faster than light — the admissibility constraints on Conf_{adm} are local in the topological substrate M , and no information is transmitted superluminally. But Phase Theory is globally nonlocal in the sense that topological invariants are global properties: the winding number of a composite configuration is defined over all of M , not at any local point. This global character is not a dynamical nonlocality (which would violate special relativity) but a topological nonlocality (which is a feature of every topologically nontrivial space). The apparent nonlocality of quantum correlations is the nonlocality of topology, which has been present in physics since Gauss's study of linking numbers and does not require revision of relativity theory.

17. Comparison with Pilot-Wave Theory and Phase Theory's Distinction

Bohmian mechanics, also known as pilot-wave theory or de Broglie-Bohm theory, is the most developed and rigorously formulated deterministic alternative to Copenhagen quantum mechanics. In Bohmian mechanics, particles have definite positions $q(t)$ at all times, guided by a pilot wave $\psi(q, t)$ that satisfies the Schrödinger equation. The guidance equation $\dot{m}q = \hbar \text{Im}(\nabla\psi/\psi)$ determines the particle velocity from the wave function. The Born rule probability distribution $|\psi|^2$ is derived — under the quantum equilibrium hypothesis — from the assumption that the initial particle distribution is given by $|\psi(q, 0)|^2$. Bohmian mechanics is deterministic (given q_0 and ψ_0 , all future positions are determined), nonlocal (the pilot wave acts instantaneously across space), and empirically equivalent to standard quantum mechanics. It is a significant achievement and an important existence proof that deterministic quantum theories are possible.

Phase Theory superficially resembles Bohmian mechanics in being deterministic and in deriving the Born rule without taking it as a postulate. But the differences are deep and systematic. The first distinction concerns ontology: Bohmian mechanics has a dual ontology — both particle positions q and the wave function ψ are fundamental entities — while Phase Theory has a single ontology: Φ alone. In Bohmian mechanics, particles are separate from the wave function; they are point masses at definite positions that the wave guides. In Phase Theory, particles are topological features of Φ itself — they have no existence separate from the phase configuration. There is no Phase Theory analog of the particle position variable q , because what appears as "particle position" is an emergent property of the topological structure of Φ in the vicinity of a region of concentrated phase winding.

The second distinction concerns the quantum equilibrium hypothesis. Bohmian mechanics derives the Born rule from the assumption that the initial

distribution of particle positions is given by $\rho_0(q) = |\psi_0(q)|^2$. This is a contingent initial condition — it must be posited, and while there are arguments (based on typicality and Dürr, Goldstein, and Zanghì's work) that $|\psi|^2$ is the "natural" initial distribution, these arguments have been criticized as circular or as invoking a measure on initial conditions whose justification is no clearer than the Born rule itself. Phase Theory derives the Born rule from the symplectic geometry of Conf_adm without any initial condition hypothesis: the measure μ_{adm} is the natural geometric measure, not an assumed initial distribution.

The third distinction concerns nonlocality. Bohmian mechanics is explicitly nonlocal: the pilot wave $\psi(q_1, q_2, \dots, q_N, t)$ is a function on the $3N$ -dimensional configuration space of N particles, and the guidance equation for particle i at position q_i depends instantaneously on the positions of all other particles through the pilot wave. This is a dynamical nonlocality — the velocity of one particle is instantaneously affected by the positions of spatially distant particles. Phase Theory has no dynamical nonlocality; its apparent nonlocality is topological. The fourth distinction concerns the wave function's ontological status: in Bohmian mechanics, ψ is a real physical field (it guides particles), while in Phase Theory, the effective quantum mechanical wave function ψ is a coarse-grained summary of the phase-geometric measure on the admissible ensemble — a tool for computing probabilities, not a fundamental field. The fifth distinction concerns the preferred basis problem: Bohmian mechanics selects the position basis as fundamental (particles have definite positions, not definite momenta or spin components), which requires justification; Phase Theory selects no preferred basis, because the measurement basis is determined by the apparatus coupling — the projection π — which is itself determined by the physics of the measurement interaction.

18. Decoherence, Environmental Coupling, and Classical Limit

Decoherence — the suppression of quantum coherence by coupling to environmental degrees of freedom — is widely regarded as the most physically well-grounded approach to the emergence of classical behavior from quantum mechanics. Pioneered by Zeh, Zurek, and their collaborators, the decoherence program explains why macroscopic objects appear to be in definite classical states (not quantum superpositions) by showing that the off-diagonal elements of the density matrix in the pointer basis decay rapidly when the system is coupled to a large environment. Decoherence is experimentally well-confirmed and theoretically robust. Phase Theory provides a structural account of decoherence that is more fundamental than the density matrix description, deriving it from the phase-geometric measure on the admissible ensemble.

In Phase Theory, define the environment E as the totality of phase degrees of freedom in Conf_adm not accessible to the measurement apparatus — the rest of the universe excluding the system S and apparatus A . The admissible ensemble for the composite system $S+E$ is $E(P, E) = \{\Phi_{\{S+E\}} \in \text{Conf_adm} \mid \Phi_{\{S+E\}} \text{ satisfies } C_P\}$. The reduced admissible ensemble for S alone is obtained by projecting out the E degrees of freedom: $E_S = \pi_S(E(P, E))$ where $\pi_S : \text{Conf_adm}^{\{S+E\}} \rightarrow \text{Conf_adm}^S$ is the projection onto the S degrees of freedom. The reduced phase-geometric measure μ_S on E_S is the marginal of μ_{adm} with respect to π_S .

The key effect of environmental coupling on the reduced measure μ_S is the suppression of sector cross-terms. When S couples to E , the composite admissible ensemble $E(P, E)$ includes configurations in which the S and E sectors are correlated. Projecting onto E_S — tracing out the E degrees of freedom — averages over the E sector contributions. If the E sector is in a superposition-like state across many different values (which is generically the

case for a macroscopic environment with many degrees of freedom), this averaging produces destructive interference between the sector boundary contributions in the S sector. Define the decoherence timescale:

$$\tau_{\text{dec}} = \kappa / (g \cdot \rho_E) \quad (18.1)$$

where g is the coupling strength between S and E, κ is the phase coherence length, and ρ_E is the environmental phase density (the number of accessible E-sector configurations per unit phase-space volume). For $\tau \gg \tau_{\text{dec}}$, the sector boundary contributions in the S reduced ensemble are completely suppressed, and the reduced measure μ_S becomes diagonal — it assigns measure only to pure sector states, with no cross-term contributions. This is the Phase Theory analog of the diagonalization of the density matrix in the pointer basis: the off-diagonal elements correspond to sector cross-terms (interference terms), and their suppression under decoherence corresponds to the Phase Theory structural result that environmental tracing eliminates boundary contributions.

The classical limit emerges naturally in this framework. When $\tau \gg \tau_{\text{dec}}$ (strong environmental coupling) and the relevant topological sector complexity is low (low winding number $\nu \approx 0$, corresponding to macroscopic objects with many particles and no long-range quantum coherence), the reduced measure μ_S on E_S is indistinguishable from a classical probability measure on a classical phase space. The symplectic volume measure of the Phase Theory sector structure reduces to the Liouville measure of classical statistical mechanics, and the emergent probabilities $P(O_j | P)$ reduce to classical Boltzmann probabilities. This is the Phase Theory derivation of the classical-quantum correspondence principle: quantum behavior is the limit of strong phase coherence (small τ_{dec}), and classical behavior is the limit of strong decoherence (large τ_{dec}). Both regimes are aspects of the same underlying phase-geometric structure, and the transition between them is

controlled by the single parameter τ_{dec} / τ — a dimensionless measure of the degree of environmental decoherence.

A notable feature of the Phase Theory account of decoherence is that it does not require the "preferred basis problem" to be solved separately. In the standard decoherence program, one must choose the basis in which the density matrix becomes diagonal — the pointer basis — and this choice is not uniquely determined by the mathematical formalism without additional physical input (the structure of the system-environment interaction Hamiltonian). In Phase Theory, the preferred basis for decoherence is determined by the projection Π_S : the sectors that survive decoherence are those that are orthogonal (in the sense of having disjoint sector boundaries) with respect to the S-E coupling. This is determined by the admissibility constraints on the composite $\text{Conf}_{\text{adm}}^{\{S+E\}}$, not by an arbitrary choice. The pointer basis is physical, not conventional.

19. Statistical Mechanics as a Special Case

Classical statistical mechanics is the most successful application of probabilistic reasoning in physics, providing rigorous derivations of thermodynamic laws from microscopic dynamics and giving quantitative predictions for transport coefficients, phase transitions, and equation-of-state properties. It would be a serious shortcoming of Phase Theory if it could not reproduce the structure of classical statistical mechanics as a special case. This section shows that it can, and that the reproduction is not an approximation but an exact structural correspondence.

In classical statistical mechanics, the canonical ensemble assigns to each microstate $(q, p) \in \Gamma$ (classical phase space) a probability proportional to $\exp(-H(q,p)/k_B T)$, where H is the Hamiltonian and T the temperature. The partition function $Z = \int_{\Gamma} \exp(-H/k_B T) dq dp$ normalizes the distribution.

The Boltzmann entropy of a macrostate M is $S = k_B \ln(\mu_{\text{Liouville}}(\Gamma_M))$ where $\Gamma_M = \{(q,p) \mid (q,p) \text{ satisfies the macroscopic constraints of } M\}$ is the phase space region corresponding to M . This structure is, in Phase Theory, derived rather than postulated in the regime where classical statistical mechanics applies.

A classical system in Phase Theory is one in which two conditions hold: (1) the topological complexity of the admissible configurations is low — specifically, only winding numbers $|n| \leq n_{\text{max}}$ are accessible, where $n_{\text{max}} \ll N$ (N being the number of degrees of freedom); and (2) the decoherence time τ_{dec} is much shorter than the dynamical time scale of interest, so that sector cross-terms are suppressed. In this regime, the admissible ensemble $E(P)$ decomposes into a classical mixture over sector states, and the phase-geometric measure μ_{adm} on $E(P)$ reduces to a measure on the space of low-winding-number configurations. The phase-coherence structure of Φ in this limit is precisely the structure of a classical field on M , and the natural measure on the space of classical field configurations is the Liouville measure.

The identification of the Phase Theory measure with the Boltzmann–Gibbs distribution proceeds as follows. The admissibility functional $I[\Phi]$ in the classical limit (low winding, strong decoherence) reduces to $I[\Phi] \approx \int_M (\kappa^2 |\nabla\Phi|^2 + H(\Phi)) d\mu_M - C$, where $H(\Phi)$ is the effective energy density of the phase configuration (the classical Hamiltonian in field-theoretic form) and C is a constant threshold. The admissibility condition $I[\Phi] \geq 0$ becomes $H[\Phi] \leq \kappa^2 \int |\nabla\Phi|^2 + C$ — an energy constraint. The phase-geometric measure μ_{adm} on configurations satisfying this constraint, in the appropriate thermodynamic limit, gives a measure proportional to $\exp(-H[\Phi]/k_B T)$ for configurations near thermal equilibrium. The Boltzmann factor emerges from the geometry of the phase-geometric measure in the classical, low-winding-

number, high-decoherence regime. Statistical mechanics is Phase Theory in the classical limit.

This unification of quantum and classical probability under a single framework — the phase-geometric measure on the admissible configuration space — is one of the most significant structural achievements of Phase Theory. It shows that the quantum-classical divide is not a fundamental ontological boundary but a regime distinction within a single, unified theory. The same measure μ_{adm} , applied to admissible ensembles in the full quantum regime, gives the Born rule; applied in the classical regime (low winding, strong decoherence), gives the Boltzmann–Gibbs distribution. Probability is the same thing in both cases — the geometric fraction of the admissible ensemble corresponding to a given outcome — and the difference in the mathematical form of the resulting distribution ($|\psi|^2$ vs. $\exp(-H/k_B T)$) reflects the difference in the topological and decoherence structure of the respective regimes.

20. Thermodynamic Arrow of Time and Phase Admissibility

The thermodynamic arrow of time — the empirical fact that entropy increases in the direction of the future and decreases toward the past — is one of the great unexplained asymmetries in physics. The fundamental dynamical laws (Newton's equations, Schrödinger equation, CPT-invariant quantum field theory) are time-symmetric or time-reversal invariant (up to CP violation, which is small). How, then, does a strongly time-asymmetric phenomenon (entropy increase) emerge from time-symmetric microdynamics? In standard statistical mechanics, the answer invokes the "past hypothesis" (Boltzmann, later formalized by Albert and Penrose): the universe began in an extraordinarily low-entropy initial state, and the second law is a consequence of this special initial condition together with the overwhelming preponderance of high-entropy future states in the microstate space. This is

an appeal to the contingent initial conditions of the universe, not to any feature of the dynamical laws themselves.

Phase Theory provides a law-based account of the thermodynamic arrow, deriving it from the asymmetry of the admissibility condition rather than from contingent initial conditions. The admissibility functional $I[\Phi]$ is not time-symmetric: it incorporates boundary conditions at both $t = t_{\text{initial}}$ and $t = t_{\text{final}}$, and these boundary conditions are not symmetric under time reversal. Specifically, the topological winding number $n[\Phi]$ has a definite value at the initial time (determined by the cosmological initial conditions, which are themselves admissible configurations — see Section 23), and the admissibility condition enforces that all subsequent configurations maintain global consistency with this initial winding number. The time-reversed evolution — starting from the final state and evolving backward — would require a different initial winding number (the time-reversed one), which is generically incompatible with the admissibility condition for the original initial conditions.

More precisely, define the set of admissible time-reversed configurations: $\text{Conf_adm}^R = \{\Phi \mid I[T[\Phi]] \geq 0\}$ where T is the time-reversal operator. For a generic admissible configuration $\Phi \in \text{Conf_adm}$, its time reversal $T[\Phi]$ is not in Conf_adm : the admissibility condition $I[\Phi] \geq 0$ is violated by $I[T[\Phi]] < 0$ for almost all Φ except for configurations of measure zero (the measure-zero set of time-symmetric configurations). This is the Phase Theory statement of the second law: the set of admissible future-directed configurations compatible with a given macrostate M at t_0 that decrease entropy at $t_1 > t_0$ has measure $\mu_{\text{adm}} = 0$. Entropy non-decrease is not a statistical tendency (overwhelmingly likely) but an admissibility theorem (zero measure for entropy-decreasing configurations).

This makes the second law exact in Phase Theory, not merely overwhelmingly probable as in standard statistical mechanics. It also explains why entropy

increase is so robust — it is not that violations are merely very unlikely, as Boltzmann's probabilistic account suggests (and which leads to the Poincaré recurrence paradox), but that they are genuinely inadmissible: they would require configurations that violate the global admissibility constraint. The past hypothesis is superseded by the admissibility structure: the universe began in a low-entropy state not because the initial conditions were fine-tuned to be special (which would require explanation), but because the admissibility constraint on cosmological boundary conditions selects for low initial entropy as a global consistency condition. The arrow of time is a theorem of Phase Theory, not an initial condition.

21. Resolution of the Measurement Problem

The measurement problem in quantum mechanics is the central unresolved difficulty of the theory. Despite decades of interpretive effort — Copenhagen, Everett, Bohmian, GRW, relational, QBist — no consensus has emerged, and each proposed resolution carries significant conceptual costs. The measurement problem has three logically distinct aspects, each of which requires independent resolution. Phase Theory resolves all three without invoking collapse, many worlds, or additional postulates, using only the structure developed in the preceding sections.

The first aspect is the problem of definite outcomes: quantum mechanics predicts that a measurement of a system in a superposition $|\psi\rangle = \alpha|+\rangle + \beta|-\rangle$ should yield a superposition of outcomes ($|\text{apparatus reads } +\rangle + |\text{apparatus reads } -\rangle$), but in fact measurements always yield definite outcomes. Phase Theory resolves this by distinguishing between the ensemble and the individual configuration. The admissible ensemble $E(P)$ for a preparation corresponding to $|\psi\rangle = \alpha|+\rangle + \beta|-\rangle$ contains configurations Φ_i in both sector E_+ (where the relevant topological observable has winding number $+1$) and sector E_- (winding number -1). But each individual Φ_i is in exactly one

sector — each physical configuration has a definite winding number, not an indefinite one. What the formalism describes as "superposition" is not a single physical configuration in an indefinite state, but the admissible ensemble $E(P)$ spanning both sectors. A measurement resolves $E(P)$ by coupling the system to the apparatus: the admissibility constraint on the composite $S+A$ system selects, for each Φ_i , the apparatus outcome compatible with the sector of Φ_i . The outcome is definite for each $\Phi_i \in E(P)$; the distribution over outcomes across the ensemble is given by the phase-geometric measure.

The second aspect is the problem of the preferred basis: what selects the eigenbasis of the measured observable as the basis in which outcomes are definite? In Phase Theory, the preferred basis is determined by the apparatus coupling, which is a physical admissibility constraint on Conf_{adm} . The measurement apparatus A is designed (physically constructed) to be sensitive to a specific topological observable of Φ — for example, the winding number in a specific direction on M corresponding to spin. The apparatus coupling defines a projection $\pi_A : \text{Conf}_{\text{adm}} \rightarrow O_A$ that is sensitive to precisely the sector boundaries corresponding to the eigenstates of the measured observable. This is not a selection of the preferred basis from "outside" the formalism — it is determined by the physical construction of the apparatus, which is itself a phase-theoretic system subject to admissibility constraints. The apparatus defines the basis by defining the projection, and the projection is determined by the admissibility-constrained coupling.

The third aspect is the problem of collapse: what physical process produces the post-measurement state? In standard quantum mechanics, collapse (the projection postulate) is a discontinuous, non-unitary update of the wave function that has no dynamical explanation within the unitary evolution described by the Schrödinger equation. Phase Theory has no collapse and no wave function, so the problem does not arise in its original form. The post-measurement state in Phase Theory is the conditioned admissible ensemble

$E(P \mid O_j) = \{\Phi_i \in E(P) \mid \pi(\Phi_i) = O_j\}$ — the subset of the original ensemble that projects to the observed outcome. This is standard Bayesian conditioning, a purely informational operation: upon receiving the measurement result O_j , the admissible ensemble is restricted to those configurations consistent with O_j . This is not a physical change in the phase configuration Φ — Φ was always in sector E_j (for the specific Φ_i that was actually realized); it is an update of the observer's representation of the admissible ensemble in light of new information. Collapse is information update, not physical state change.

The resolution of the measurement problem in Phase Theory is more complete than any existing alternative. Copenhagen simply denies that the question has a physical answer ("quantum mechanics describes observations, not reality"). Everett's many-worlds interpretation answers the first and second aspects but struggles with the third (the status of probability in a theory where all outcomes are realized). Bohmian mechanics answers all three but introduces a preferred basis (position), a nonlocal pilot wave, and the quantum equilibrium hypothesis. GRW (Ghirardi-Rimini-Weber) collapse models introduce stochastic collapse as a new fundamental process. Phase Theory answers all three without any of these costs: no preferred basis, no nonlocal dynamics, no stochastic collapse, no many worlds, and no additional postulates beyond those already in the theory.

The key insight of the Phase Theory resolution is the distinction between the ensemble and the individual configuration. Standard quantum mechanics conflates these: the wave function $|\psi\rangle$ is simultaneously a description of an individual quantum system and a probability amplitude for measurement outcomes. This dual role is the source of the measurement problem — the wave function cannot simultaneously be an individual physical state (which should evolve unitarily) and a probability amplitude (which should collapse on measurement). Phase Theory separates these roles cleanly: the admissible

ensemble $E(P)$ is the probability-relevant object (from which the Born rule measure is derived), while individual configurations $\Phi_i \in E(P)$ are the physical states (each fully determinate and unitarily — or more precisely, admissibility-preservingly — evolving). The apparent paradox of a system being both in superposition and yielding definite outcomes is resolved by recognizing that "in superposition" describes the ensemble, while "yields a definite outcome" describes each individual configuration.

22. Resolution of Foundational Tensions in Quantum Foundations

Beyond the measurement problem, quantum mechanics exhibits a cluster of foundational tensions — apparent contradictions or unresolved conceptual puzzles — that have generated sustained philosophical and physical debate since the theory's inception. Phase Theory systematically addresses each of these tensions, resolving them through the structural account of probability, measurement, and dynamics developed in the preceding sections. This section presents five major tensions and their Phase Theory resolutions in a systematic format.

Tension 1: Randomness versus determinism. Standard quantum mechanics is dynamically deterministic (the Schrödinger equation is deterministic) but measurably indeterministic (measurement outcomes are random). This combination is philosophically unstable: if the dynamics is deterministic, what is the source of the indeterminism? Phase Theory resolves this by establishing that both sides of the tension are correctly described but require reinterpretation. The dynamics is indeed deterministic — the admissibility-preserving evolution of Φ is fully determined, as established in Section 11. The apparent indeterminism of measurement outcomes is real but not ontological: it reflects the genuine underdetermination of Φ by the preparation procedure P (the admissible ensemble $E(P)$ contains many Φ_i)

combined with the coarse-graining of the outcome space O by the projection π . Both facts are structural features of Phase Theory, and both are consistent with full ontological determinism. There is no tension.

Tension 2: The observer's privileged role. In Copenhagen quantum mechanics, the "observer" plays a logically indispensable but physically mysterious role: the wave function evolves unitarily until an observer makes a measurement, at which point it collapses. The observer is not subject to the Schrödinger equation in the same way that the observed system is. This asymmetric treatment of observers and physical systems is widely regarded as a fundamental defect of the Copenhagen interpretation. Phase Theory resolves this by making the apparatus a phase-theoretic system subject to the same admissibility constraints as the measured system. There is no privileged observer, no special role for consciousness or subjective experience, and no asymmetric treatment of measuring devices and physical systems. The projection π is determined by the physical coupling between the apparatus and the system — a purely physical, admissibility-constrained interaction. The observer is demystified: it is a macroscopic phase-theoretic system whose coupling to the measured system defines a projection on Conf_adm .

Tension 3: Non-locality. The apparent nonlocality of quantum mechanics — manifest in EPR correlations, Bell inequality violations, and the instantaneous action of wave function collapse at a distance — is in tension with special relativity's prohibition on faster-than-light causal influences. Phase Theory resolves this by distinguishing topological nonlocality (the global character of winding numbers and other topological invariants) from dynamical nonlocality (faster-than-light signal transmission). Topological invariants are global properties of the phase configuration by definition — a winding number is the integral of a form over a closed manifold, not a local quantity. The fact that measurements at spacelike-separated locations are correlated through the global topology of Φ does not involve any signal or influence

propagating between the locations; the correlations are established by the global admissibility constraint, which is imposed simultaneously on all of M . No causal influence is transmitted; no relativistic constraint is violated. The nonlocality of topology is compatible with relativistic locality of dynamics.

Tension 4: Wave-particle duality. Quantum objects appear to exhibit both wave-like behavior (interference, diffraction) and particle-like behavior (localization, discrete impacts on detectors). No classical concept captures both aspects simultaneously, leading to the dual description of quantum objects as waves-and-particles, which is explanatorily unsatisfying. Phase Theory resolves this by providing a single ontological account from which both aspects emerge. Particles are topological features of the phase field Φ — regions of concentrated winding where the phase field undergoes rapid topological change. Waves are propagating modes of the phase field in regions away from the topological features. Duality is resolved because both are features of the same object — the phase field Φ — in different aspects of its structure. The wave-like behavior is the dynamics of the phase modes; the particle-like behavior is the topology of the concentrated winding features. Neither is fundamental independently; both are emergent aspects of the same underlying field.

Tension 5: Ontological status of the wave function. Is the quantum wave function ψ a real physical field (ψ -ontic interpretations) or a representation of knowledge (ψ -epistemic interpretations)? The ψ -ontic view (Everett, GRW, Bohmian) faces the difficulty of explaining how an entity defined on the $3N$ -dimensional configuration space can be physically real in 3-dimensional space. The ψ -epistemic view (QBism, epistemic interpretations) faces the difficulty of explaining why different agents with the same information always have the same wave function, and why the wave function predicts with such precision. Phase Theory dissolves the dilemma: the wave function is neither straightforwardly ψ -ontic nor purely ψ -epistemic. It is an effective coarse-

grained representation of the phase-geometric measure on the admissible ensemble $E(P)$ — a mathematical encoding of the sector structure and phase amplitudes of $E(P)$ in a compact Hilbert-space notation. It is not a fundamental field (it has no independent ontological status in Phase Theory), but it is not merely subjective (it is determined by the objective geometry of Conf_adm). The wave function is a derived effective description, and its collapse is the information update of this effective description upon receiving measurement outcomes.

23. Implications for Probability in Cosmology

The application of probability to cosmological questions is among the most philosophically fraught areas of theoretical physics. Cosmology deals with unique objects and events — there is only one universe, one big bang, one set of cosmological parameters — and the standard probability interpretations all face fundamental difficulties when applied to such unique cases. Frequentist probability is undefined (there are no repeated trials of the universe). Bayesian probability requires a prior over possible universes, but the choice of prior is unconstrained by empirical data (we have access to only one universe), and different priors lead to very different anthropic conclusions. Propensity interpretations struggle with the absence of an external physical setup that could ground the propensity. Yet quantum cosmology (Hartle–Hawking no-boundary proposal, tunneling from nothing, eternal inflation) demands probabilistic reasoning about universe-scale events. Phase Theory provides a natural and well-grounded framework for cosmological probability.

In Phase Theory, the cosmological admissible ensemble is defined by: $E(\text{cosm}) = \{\Phi_i \in \text{Conf_adm} \mid I[\Phi_i] \geq 0, \text{ with cosmological boundary conditions } B_{\text{cosm}}\}$. The cosmological boundary conditions B_{cosm} are the admissibility constraints applicable at cosmological scales — conditions on

the global topology of Φ over the cosmological manifold M_{cosm} , including the total winding number of the universe and the large-scale phase coherence structure. These boundary conditions are determined by the structure of Phase Theory itself (specifically, by the admissibility functional evaluated on cosmological configurations) and are not arbitrary choices. The cosmological admissible ensemble $E(\text{cosm})$ is the set of all globally admissible phase configurations consistent with these boundary conditions — the set of all possible universe histories that are physically realizable according to Phase Theory.

The phase-geometric measure μ_{adm} on $E(\text{cosm})$ defines objective cosmological probabilities. The probability of a given universe history H (characterized by its cosmological observables — the Hubble constant, the primordial spectrum of density fluctuations, the dark energy density, and so on) is: $P(H \mid \text{cosm}) = \mu_{\text{adm}}(\{\Phi \in E(\text{cosm}) \mid \pi_{\text{cosm}}(\Phi) = H\}) / \mu_{\text{adm}}(E(\text{cosm}))$, where π_{cosm} is the projection from the cosmological admissible configuration space to the space of macroscopic cosmological observables. No external observer, no agent, and no reference class is required: the probability is intrinsic to the admissible structure of the cosmological phase configuration, determined by the geometry of Conf_{adm} evaluated on cosmological scales. This is a fully objective probability for cosmological events — something that no interpretation of standard quantum cosmology has succeeded in providing.

The anthropic principle — the observation that the cosmological parameters must be compatible with the existence of observers, and the use of this observation to constrain the probability distribution over cosmological parameters — has a natural grounding in the Phase Theory framework. In standard anthropic reasoning (particularly in the context of the string theory landscape or eternal inflation), the probability of observing a given set of cosmological parameters depends on the prior probability distribution over

possible universes (the measure on the landscape) combined with selection effects (the observer must exist in order to make observations). The measure on the landscape is notoriously ill-defined, leading to the "measure problem" in cosmology. In Phase Theory, the measure on $E(\text{cosm})$ is μ_{adm} — uniquely determined by the geometry of the cosmological admissible configuration space. Anthropic selection is incorporated by conditioning on the constraint "the universe contains observers," which restricts $E(\text{cosm})$ to configurations consistent with the formation of complex structures (stars, planets, biological life). The conditioned measure gives the Phase Theory anthropic probability distribution over cosmological parameters, resolving the measure problem without invoking arbitrary priors.

24. Implications for Information Theory and Entropy

Information theory, founded by Shannon in 1948 and extended to quantum systems by von Neumann, provides the mathematical framework for quantifying the amount of information encoded in a probability distribution. Shannon entropy $H = -\sum_i p_i \log p_i$ (for a discrete distribution with probabilities p_i) and its quantum generalization $S = -\text{Tr}(\rho \ln \rho)$ (for a density matrix ρ) are the central quantities. Since Phase Theory derives probability from phase-geometric measure, it similarly derives information-theoretic quantities from the geometry of the admissible configuration space — unifying classical and quantum information under a single structural account.

In Phase Theory, the Shannon entropy of the outcome distribution for a preparation P and projection π is: $H(P, \pi) = -\sum_j P(O_j | P) \log P(O_j | P) = -\sum_j [\mu_{\text{adm}}(E_j) / \mu_{\text{adm}}(E(P))] \log[\mu_{\text{adm}}(E_j) / \mu_{\text{adm}}(E(P))]$, where $E_j = \pi^{-1}(O_j) \cap E(P)$ is the sector of the admissible ensemble projecting to outcome O_j . This is the logarithm of the accessible phase-sector volumes, weighted by their fractional contributions. Information is not a primitive — it is the geometric measure of accessible phase configurations. H is large when the

admissible ensemble $E(P)$ is spread across many sectors of roughly equal measure (high uncertainty about outcomes); H is small when $E(P)$ is concentrated in a few sectors (high predictability). This geometric interpretation of information generalizes naturally to continuous distributions and to the quantum case.

The von Neumann entropy $S = -\text{Tr}(\rho \ln \rho)$ is similarly derived in Phase Theory. The density matrix ρ is the effective representation of the reduced phase-geometric measure μ_S on the system's admissible ensemble E_S (after tracing out the environment, as in Section 18). Specifically, $\rho_{jk} = \langle j|\rho|k \rangle = \mu_{\text{adm}}(E_j \cap E_k) / \mu_{\text{adm}}(E(P))$, where E_j and E_k are the sector subsets of E_S . The diagonal elements $\rho_{jj} = \mu_{\text{adm}}(E_j) / \mu_{\text{adm}}(E(P))$ are the Born rule probabilities; the off-diagonal elements ρ_{jk} encode the sector boundary overlaps (interference terms). The von Neumann entropy $S = -\text{Tr}(\rho \ln \rho) = -\sum_j \lambda_j \log \lambda_j$ (where λ_j are the eigenvalues of ρ) is the logarithm of the accessible sector volume in the reduced configuration space, measured by the eigenvalues of the reduced measure operator. The equivalence of Shannon and von Neumann entropy in Phase Theory — both are the logarithm of accessible phase-sector volume — establishes that classical and quantum information are the same structural entity in different topological regimes.

The black hole information paradox — the question of whether information is destroyed when matter falls into a black hole and the black hole subsequently evaporates via Hawking radiation — takes on a new character in the Phase Theory framework. The paradox arises because Hawking radiation appears to be thermal (carrying no information about the infalling matter), but unitarity of quantum mechanics implies that information cannot be destroyed. In Phase Theory, information is identified with the logarithm of the accessible phase-sector volume in Conf_{adm} . The question becomes: is the phase-geometric measure μ_{adm} on Conf_{adm} globally conserved under the admissibility-preserving evolution, including the evolution in the presence of black holes?

The global consistency enforcement (Axiom PT3) implies that μ_{adm} is globally conserved — the admissibility-preserving evolution of a phase configuration is, in its abstract form, a flow on Conf_{adm} that preserves the symplectic structure ω and hence the Liouville measure μ_{adm} . Therefore, the phase-geometric measure of the accessible sector is not destroyed by black hole formation and evaporation; it is merely redistributed across Conf_{adm} in a way that appears thermal to observers with limited access to the full configuration. The information paradox is resolved: information is globally conserved by the Phase Theory analog of unitarity (admissibility-preservation), and the apparent information loss is a consequence of the coarse-graining of the Hawking radiation by the external observer's projection.

25. Experimental and Falsifiable Consequences

A theoretical framework that cannot be distinguished from its competitors by experiment is not a contribution to physics, however philosophically elegant it may be. Phase Theory, including its account of emergent probability, makes specific, quantitatively precise, falsifiable predictions that differ from those of standard quantum mechanics and from those of other quantum foundations programs. Five concrete predictions are described here, each with a detection protocol and an order-of-magnitude estimate of the expected effect.

Prediction 1 — Born rule corrections in topologically nontrivial media. The derivation of the Born rule in Section 9 assumes that the admissible configuration space Conf_{adm} has the standard symplectic structure of the phase field on a topologically trivial substrate M . In topologically nontrivial materials — crystalline systems with non-trivial band topology (topological insulators, Weyl semimetals, axion insulators) — the effective admissible configuration space for electronic degrees of freedom has a modified symplectic structure, reflecting the non-trivial topology of the Brillouin zone.

This modification produces small but in-principle-detectable corrections to the Born rule probabilities for quantum measurements conducted in such materials. The correction scales as $\delta P \sim (\kappa/L)^2$, where κ is the phase coherence length of the electronic system and L is the system size. For a topological insulator with $\kappa \sim 10$ nm and $L \sim 1$ mm, this gives $\delta P \sim 10^{-10}$ — far below current precision. However, for mesoscopic samples with $L \sim 100$ nm, the correction $\delta P \sim 10^{-2}$ falls within the sensitivity of precision interferometric measurements. Detection protocol: compare the visibility of quantum interference fringes in a mesoscopic topological insulator ring with the Born rule prediction for a trivially topological reference material of the same composition. A systematic enhancement of visibility by δP above the Born rule baseline in the topological sample would be a positive signal.

Prediction 2 — Super-Poissonian suppression in macroscopic phase-coherent systems. In Bose–Einstein condensates (BECs) and superconductors, the admissible ensemble has dramatically reduced entropy: the many-body phase configuration collapses into a single coherent sector (the condensate or superconducting ground state), reducing $E(P)$ to a much smaller set than would be expected for an uncorrelated N -body system. Phase Theory predicts that, in measurement of appropriate observables in such systems, the variance of the outcome distribution is reduced below the Born rule expectation for uncorrelated particles — a super-Poissonian reduction in outcome variance. This corresponds, in quantum optical language, to a number-squeezed state; but Phase Theory predicts this reduction as a generic consequence of sector condensation in the admissible ensemble, not merely as a special property of specific squeezed states. Detection protocol: photon counting statistics in a Bose–Einstein condensate coupled to an optical cavity; comparison of the Mandel Q parameter with the Phase Theory prediction as a function of condensate fraction.

Prediction 3 — Topology-dependent modification of the Tsirelson bound. The Tsirelson bound $B \leq 2\sqrt{2}$ on Bell correlations is, in standard quantum mechanics, a consequence of the Hilbert space structure of quantum states. In Phase Theory, the bound is determined by the symplectic geometry of the composite admissible ensemble — specifically, by the phase-geometric measure on the sector boundaries of the two-particle configuration space. For systems with non-trivial topological invariants — anyonic systems in fractional quantum Hall states, Majorana modes in topological superconductors — the topological invariant ν of the system modifies the effective phase-geometric measure, producing a modification of the Tsirelson bound: $B \leq 2\sqrt{2} \cdot f(\nu)$, where $f(\nu)$ is a computable function of the topological invariant satisfying $f(0) = 1$ (recovering the standard bound for trivially topological systems) and $f(1) > 1$ for non-trivial topology. For a system with $\nu = 1/2$ (as in a $\nu = 1/2$ fractional quantum Hall state), Phase Theory predicts a modification of order $f(1/2) \approx 1.05$, giving an enhanced Tsirelson bound $B \leq 2\sqrt{2} \cdot 1.05 \approx 2.97$. Detection protocol: Bell inequality test with anyonic quasi-particles in a fractional quantum Hall system, comparing the maximum observed Bell violation with the standard and modified Tsirelson bounds.

Prediction 4 — Non-Markovian memory effects in ensemble evolution. In standard quantum mechanics, a system coupled to a Markovian environment evolves according to the Lindblad master equation, which describes a Markov process: the future state of the system depends only on its current state, not on its history. Phase Theory, being deterministic at the level of the full phase configuration, predicts non-Markovian evolution of the probability distribution $P(O_j, t)$: the probability at time t_3 is correlated with the probability at t_1 in a way that cannot be explained by a Markov chain through t_2 (with $t_1 < t_2 < t_3$). This non-Markovianity is not due to memory in the system-environment coupling (which can produce non-Markovian behavior in standard quantum mechanics for structured environments) but to the deterministic admissibility-preserving evolution of the full phase

configuration, which produces correlations across time that exceed those possible in any Markov chain. Detection protocol: measurement of the non-Markovianity measure N defined by Breuer, Laine, and Piilo (based on the rate of change of trace distance between pairs of initial states) for a system with long phase coherence, comparing the observed N with the Phase Theory prediction versus the Markovian Lindblad prediction.

Prediction 5 — Gravitational correction to the decoherence rate. Standard quantum mechanics, when combined with Newtonian gravity (Diósi-Penrose model) or linearized general relativity, predicts a gravitationally-induced decoherence rate $\tau_{\text{grav}}^{-1} \sim G M^2 / (\hbar r)$, where G is Newton's constant, M is the mass of the superposed object, and r is the spatial separation of the superposed wave packets. Phase Theory modifies this prediction because gravity, in Phase Theory, is the emergent geometry of the phase coherence structure of Φ (established in Paper 3), which modifies the effective admissible configuration space Conf_{adm} in the presence of gravitational fields. The Phase Theory decoherence rate is: $\tau_{\text{dec}}^{-1}(g) = \tau_{\text{dec}}^{-1}(0) \cdot (1 + \alpha G M / r c^2)$, where $\tau_{\text{dec}}(0)$ is the decoherence rate in the absence of gravity and α is a dimensionless Phase Theory parameter of order unity, calculable from the admissibility functional. The multiplicative correction factor $(1 + \alpha G M / r c^2)$ is topology-dependent and differs from the additive correction that would be expected from a straightforward combination of the Diósi-Penrose model with Phase Theory. Detection protocol: matter-wave interferometry of large molecules or nanospheres in a controlled gravitational potential, measuring the decoherence rate as a function of gravitational potential energy Mgr , and comparing the measured functional form with the Diósi-Penrose prediction and the Phase Theory prediction.

26. Relation to Paper 6 — Collapse-Free Measurement Theory

The present paper has established the foundational account of probability within Phase Theory: probability is not primitive, but emerges as the phase-geometric measure of admissible phase sectors under coarse-grained access and environmental coupling. The Born rule has been derived (Section 9), the measurement problem has been resolved at the level of outcomes and the preferred basis (Section 21), and post-measurement conditioning has been identified with standard information-theoretic updating of the admissible ensemble. What remains is the dynamical account of the measurement interaction itself — the detailed physical description of how the coupling between a measuring apparatus A and a system S implements the projection π and produces the conditioned admissible ensemble $E(P \mid O_j)$. This is the subject of Paper 6 of the Phase Theory series, "Measurement Without Collapse."

The present paper treats measurement as a projection $\pi : \text{Conf_adm} \rightarrow O$ and characterizes the post-measurement ensemble as $E(P \mid O_j) = \{\Phi \in E(P) \mid \pi(\Phi) = O_j\}$. This is the kinematic description of measurement — what the measurement does to the admissible ensemble. Paper 6 provides the dynamical description — how the admissibility constraints on the composite $S+A$ system evolve over the measurement interaction time Δt to implement the projection π . The key result of Paper 6 is that the joint admissible ensemble $E(S+A)$ factorizes after the measurement interaction into a sum of apparatus-reading sectors: $E(S+A) = \sqcup_j E(S+A, j)$, where $E(S+A, j) = \{\Phi_{\{S+A\}} \in E(P_{\{S+A\}}) \mid \Phi_A \text{ reads outcome } j\}$. The weights of these sectors in the joint phase-geometric measure are exactly the Born rule probabilities $\mu_{\text{adm}}(E(S+A, j)) = |A_j|^2 \cdot \mu_{\text{adm}}(E(P_{\{S+A\}}))$, using the result of Theorem 9.1. This factorization — the Phase Theory account of the unitary pre-measurement evolution followed by effective "collapse" to a definite outcome — uses no new concepts: it is the application of the admissibility

constraints, ensemble formation, and phase-geometric measure developed in Papers 1–5 to the specific case of the measurement interaction.

The significance of this relationship between Papers 5 and 6 is that it demonstrates the self-sufficiency of the Phase Theory framework. The entire quantum mechanical formalism — state preparation, unitary evolution, measurement, Born rule probabilities, post-measurement state — is derived from the three axioms of Phase Theory (PT1–PT3) without any additional structure. Paper 5 (the present paper) establishes the probability theory; Paper 6 completes the measurement theory; together, they constitute a full, collapse-free, deterministic account of quantum mechanics that is more fundamental than the standard formalism and free of its foundational difficulties. The eight papers of the Phase Theory series — on the ontological foundations, the emergence of spacetime, the emergence of the standard model, the emergence of quantum field theory, the emergence of probability (the present paper), the measurement theory, the cosmological implications, and the experimental program — form a complete and internally consistent theoretical structure.

27. Philosophical Implications — Determinism, Agency, and Knowledge

Phase Theory recovers a form of determinism that has not previously been available in the philosophy of physics. It is not Laplacian determinism, which posits that a demon with complete knowledge of all positions and momenta at a time t_0 can predict all future positions and momenta. Laplacian determinism is compatible with quantum mechanics only if quantum mechanics is interpreted as an epistemic theory (QBism, epistemic hidden variables), which is no longer defensible given Bell's theorem. Phase Theory's determinism is what we may call structural determinism: the physical world evolves according to the global admissibility constraint on Φ , which is fully

deterministic, but no physical subsystem can access the full information needed to predict outcomes beyond ensemble statistics. The determinism is real — Nature does not flip coins — but the predictability is structurally limited. This combination is unprecedented: previous deterministic theories (classical mechanics, Bohmian mechanics) allowed in-principle complete predictability; previous indeterministic theories (Copenhagen, GRW) denied that the evolution was fully determined. Phase Theory is deterministic in a way that is epistemically bounded not by computational limits but by structural admissibility constraints.

The implications for scientific epistemology are significant. The limits of predictability in Phase Theory are not contingent features of the current state of science — features that might be overcome by a more powerful theory, more precise instruments, or greater computational resources. They are permanent structural features of the phase-topological world: any physical measuring apparatus is a phase-theoretic system subject to admissibility constraints, and these constraints prevent any apparatus from accessing the full phase configuration without becoming as large as the universe. The lesson for science is not pessimism — Phase Theory makes more precise and more fundamental predictions than standard quantum mechanics in many regimes — but intellectual humility: the ensemble statistics are the best possible prediction, not because of contingent limitations, but because of the structure of reality. Science is the study of admissible ensembles; individual events are epistemically inaccessible, not because of any failure of science, but because access to individual events is itself admissibility-constrained.

The questions of agency and free will take on new content in the context of Phase Theory's structural determinism. An agent — a human being, a robot, a decision-making system — is, in Phase Theory, a phase-theoretic system: a configuration $\Phi_A \in \text{Conf_adm}$ with sufficient topological complexity to implement the functional organization associated with agency (goal-

directedness, deliberation, learning). The "decisions" of an agent are phase-theoretic events — admissibility-constrained evolutions of Φ_A in Conf_{adm} — and they are, at the level of the full phase configuration, fully determined by the global admissibility constraint. This is hard determinism in the ontological sense: the future "choices" of an agent are, in principle, determined by the full phase configuration at any earlier time. However, since no physical system — including the agent itself and any external observer of the agent — can access the full phase configuration, the agent's future decisions are effectively unpredictable, even to the agent. This effective unpredictability is the structural foundation for the phenomenology of deliberation and choice: deliberation is the process by which an agent explores the sector structure of its own admissible ensemble, seeking configurations with desired topological properties. Compatibilist free will — the sense in which deliberated choices are genuinely the agent's own, even in a deterministic world — is fully supported by Phase Theory. Libertarian free will — the sense in which the agent's choices are not determined by prior causes — is refuted, but this refutation is a structural theorem of Phase Theory, not a philosophical preference.

The epistemological implications extend to the status of theoretical physics itself. Physical theories, in the Phase Theory framework, are coarse-grained descriptions of admissible ensembles — effective representations of the phase-geometric measure on $E(P)$ for various preparations P and projections π . Standard quantum mechanics is an effective description valid in the regime where the full Phase Theory structure can be approximated by a Hilbert space with unitary dynamics and Born rule measurement. Classical mechanics is an effective description valid in the classical limit (low winding, strong decoherence). General relativity is an effective description of the large-scale phase coherence geometry. No effective description is fundamental; all are approximations to the full Phase Theory structure. But the approximations are excellent in their respective regimes of validity, which is why they have

been so empirically successful. Phase Theory unifies these effective descriptions under a single structural account, providing the framework within which the limits of each approximation can be calculated and the corrections beyond those limits can be predicted.

28. Comparison with Other Quantum Foundations Programs

The landscape of quantum foundations programs is rich and contentious. Each major program has achieved important results while facing significant unresolved difficulties. A systematic comparison of Phase Theory with the major programs clarifies Phase Theory's contribution and its relationship to the existing literature. We examine five programs in detail: the Everett (Many Worlds) interpretation, Relational Quantum Mechanics (RQM), QBism, Consistent Histories, and the quantum gravity programs (Loop Quantum Gravity and String Theory).

The Everett interpretation, or Many Worlds interpretation (Everett 1957; DeWitt 1967; Deutsch 1985; Wallace 2012), retains the wave function as a fundamental ontological entity and denies the projection postulate: all branches of the wave function are real, and measurement merely entangles the observer with the system in a larger wave function. This approach has the virtue of eliminating collapse while preserving unitarity. Its primary difficulty is the probability problem: in a theory where all branches are real, what grounds the Born rule probabilities? Deutsch (1999) and Wallace (2012) developed a decision-theoretic derivation of the Born rule from Everettian axioms, arguing that a rational agent in a branching universe should assign credences proportional to $|\psi|^2$ for self-interested reasons. This derivation has been criticized as circular (it assumes rational agents can act as if the Born rule holds, which is what needs to be explained) and as question-begging (the decision-theoretic framework presupposes a notion of probability that it then purports to derive). Phase Theory resolves the probability problem without

any appeal to rational agency or decision theory: the Born rule is a theorem about the symplectic geometry of Conf_adm , independent of any notion of rational belief. Furthermore, Phase Theory has no branching — there is one universe, one phase configuration, and no proliferation of ontological entities.

Relational Quantum Mechanics (Rovelli 1996; Laudisa and Rovelli 2019) holds that quantum states are not absolute but relative to a reference system: the quantum state of S relative to O encodes O 's information about S . RQM dissolves the measurement problem by making quantum states perspectival — the collapse of the wave function relative to O is simply O 's acquisition of information about S . This approach is compatible with Phase Theory's account of contextuality (measurement outcomes depend on the apparatus projection π), but it differs from Phase Theory in an important respect: RQM retains probability as primitive and relational, while Phase Theory derives probability from the objective geometry of Conf_adm . In RQM, probabilities are the credences of observers relative to reference systems; in Phase Theory, probabilities are the objective phase-geometric fractions of the admissible ensemble. This distinction matters for cosmological applications: RQM faces difficulties in applying its relational framework to the universe as a whole (there is no external reference system), while Phase Theory's objective phase-geometric measure applies to the cosmological admissible ensemble without requiring an external observer.

QBism (Fuchs and Schack 2013; Fuchs, Mermin, and Schack 2014) treats quantum mechanics as a user's manual for an agent to update her beliefs about the consequences of her actions on the world. The quantum state is an agent's belief state; the Born rule is a normative constraint on how to update beliefs in light of measurement outcomes. QBism is empirically adequate (it makes no predictions that differ from standard quantum mechanics) and philosophically sophisticated (it dissolves certain paradoxes by relocating them from physics to epistemology). Phase Theory agrees with QBism that

quantum states encode information states — the effective description of the phase-geometric measure on $E(P)$ — but provides the objective physical ground for those information states that QBism deliberately denies. For QBism, there is no deeper story about why the Born rule constrains rational belief in the way it does; for Phase Theory, the story is the symplectic geometry of Conf_adm . Phase Theory is therefore more explanatory than QBism: it answers the question QBism refuses to ask.

The Consistent Histories (or Decoherent Histories) approach (Griffiths 1984; Omnès 1988; Gell-Mann and Hartle 1990) assigns probabilities to coarse-grained histories of quantum systems, subject to consistency conditions (decoherence conditions) that ensure the classical probability calculus applies to the chosen family of histories. This approach is structurally similar to Phase Theory's ensemble-and-projection framework: coarse-grained histories correspond to sequences of projections on Conf_adm , and the consistency conditions correspond to the requirement that sector boundaries do not overlap (no cross-terms) between different histories in the chosen family. The key difference is that Consistent Histories retains probability as primitive: the consistency conditions are formulated in terms of density matrix traces (quantum mechanical probabilities), and the framework provides conditions under which those probabilities satisfy classical rules, without explaining why the probabilities take the values they do. Phase Theory derives the consistency conditions from admissibility constraints: two histories are decoherent (consistent) in Phase Theory precisely when their corresponding sector boundaries in Conf_adm do not overlap — a geometric condition on Conf_adm that requires no probabilistic primitive to state.

Loop Quantum Gravity (LQG) and String Theory are not quantum foundations programs but quantum gravity theories; however, they make implicit commitments to the probabilistic structure of quantum mechanics that Phase Theory's account challenges. LQG, in its canonical formulation (Rovelli and

Smolin 1995; Thiemann 2007), quantizes the gravitational field using standard quantum mechanical methods, including the Hilbert space structure and the Born rule as postulates. String theory similarly assumes standard quantum mechanics (with modifications at the Planck scale) and uses the Born rule to compute scattering amplitudes. Phase Theory is potentially compatible with both programs at the level of effective descriptions: LQG's spin networks and spin foams may be interpretable as coarse-grained representations of the topological structure of Conf_adm on cosmological scales, and string amplitudes may be reproduced by the phase-geometric measure on the string theory sector of Conf_adm . The key contribution of Phase Theory to these programs is the replacement of the Born rule postulate with a derived theorem: the quantum mechanical probabilities computed in LQG and string theory are not fundamental features of those theories but emergent geometric quantities derived from the phase-geometric measure on the admissible configuration spaces of the respective theories.

29. Mathematical Summary — Theorems and Propositions

This section compiles the principal formal results of the paper in a consolidated mathematical summary. Each result is stated in the form of a numbered proposition, theorem, or corollary, with a brief indication of the section in which the proof appears or a sketch of the proof. The results are organized in order of logical dependence: each result follows from those preceding it.

Proposition 29.1 (Phase Space is Not a Probability Space).

The admissible configuration space Conf_adm equipped with the phase-geometric measure μ_adm is not a probability space in the sense of Kolmogorov, because $\mu_adm(\text{Conf_adm})$ is not normalized to unity. It is a measure space

(Conf_adm, $\mathcal{B}$, μ_{adm}) where $\mathcal{B}$ is the Borel σ -algebra induced by the phase topology. Probability spaces arise only after conditioning on a preparation P : the conditional space $(E(P), \mathcal{B}|_{\{E(P)\}}, \mu_{\text{adm}}|_{\{E(P)\}} / \mu_{\text{adm}}(E(P)))$ is a probability space.

This proposition clarifies the logical status of the Phase Theory measure. The global space Conf_adm is an unnormalized geometric measure space; probability spaces are derived by conditioning on physically realized preparations. The proposition is established by the definitions of Section 4 (Conf_adm and μ_{adm}) and Section 7 (the derivation of the probability space structure by conditioning on P).

Theorem 29.2 (Emergence of Kolmogorov Probability).

For any preparation P and projection π , the function $P(O_j | P) = \mu_{\text{adm}}(\{\Phi \in E(P) | \pi(\Phi) = O_j\}) / \mu_{\text{adm}}(E(P))$ defines a probability measure on the outcome space O satisfying all three Kolmogorov axioms: (i) non-negativity, (ii) normalization, and (iii) countable additivity. No probability primitive is invoked in the proof.

Proved in Section 7 (Theorem 7.1) using the measure-theoretic properties of μ_{adm} (non-negativity, σ -additivity) and the partition property of the coarse-graining projection π established in the Coarse-Graining Theorem (6.1). The proof is constructive: it exhibits the Kolmogorov structure as a consequence of the geometric structure of Conf_adm.

Theorem 29.3 (Born Rule).

For an admissible ensemble $E(P)$ with sector decomposition $E(P) = \sqcup_k E_k$, where E_k are the topological sectors of Φ with phase amplitudes A_k , the phase-geometric measure satisfies $\mu_{\text{adm}}(E_k) = |A_k|^2 \cdot \mu_{\text{adm}}(E(P))$. Therefore $P(k | P) = |A_k|^2$. The Born rule is a theorem of Phase Theory.

Proved in Section 9 (Theorem 9.1) from the symplectic volume computation of Section 8 and Appendix B. The key steps are: (1) restriction of the symplectic form ω to the amplitude space $\{A_k\}$; (2) computation of the Liouville volume $\text{Vol}(E_k) = |A_k|^2 \cdot \text{Vol}(E(P))$ from the Fubini–Study-like metric on the phase amplitude space; (3) normalization to give the Born rule probability.

Corollary 29.4 (No Stochastic Primitive).

The derivation of Theorems 29.2 and 29.3 uses no stochastic axiom, no Born rule postulate, no collapse postulate, and no random variable. Probability is entirely derived from the deterministic admissibility structure of Phase Theory.

This corollary follows immediately from inspection of the proofs: every step in the derivations of Theorems 29.2 and 29.3 uses only (a) the definition of the admissible configuration space Conf_{adm} (Section 4), (b) the natural symplectic measure μ_{adm} (Section 8), (c) the topological sector decomposition (Sections 4 and 9), and (d) standard measure theory (σ -additivity of measures). No probabilistic axiom appears at any step.

Theorem 29.5 (Structural Unpredictability).

For any physical agent A modeled as a phase-theoretic subsystem of Conf_{adm} , the measurement interaction between A and a system S constitutes an admissibility constraint on the composite admissible ensemble $E(A+S)$ that prevents A from accessing the full configuration Φ of S . Therefore, A cannot predict the outcomes of measurements on S beyond the ensemble probabilities $P(O_j | P)$. Unpredictability is a theorem, not an assumption.

Proved in Section 11 (Theorem 11.1) by demonstrating that complete access to Φ_S by any physical agent A would require A to correlate with the full global phase configuration Φ_{total} , which is possible only if $A =$ the entire universe — a

contradiction since A is a proper subsystem. The proof uses the global character of the admissibility constraint and the compactness of M .

Theorem 29.6 (Classical Limit).

In the limit of strong environmental coupling ($g \rightarrow \infty$) and low topological complexity ($\nu \rightarrow 0$), the phase-geometric measure μ_{adm} reduces to the Liouville measure on classical phase space, and the emergent probabilities $P(O_j | P)$ reduce to classical statistical mechanical probabilities. Classical and quantum probability are the same structure in different regimes.

Established in Sections 18 and 19. The key steps are: (1) In the strong-coupling limit ($g \rightarrow \infty$), decoherence time $\tau_{\text{dec}} \rightarrow 0$, and sector cross-terms in the reduced measure μ_S are fully suppressed. (2) In the low-complexity limit ($\nu \rightarrow 0$), only low-winding-number sectors contribute to the admissible ensemble. (3) In this combined limit, the sector decomposition of Conf_{adm} reduces to the classical phase space partition, and μ_{adm} reduces to the Liouville measure. (4) The Born rule probabilities $P(k | P) = |A_k|^2$ reduce to the Boltzmann probabilities $p_k = \exp(-E_k/k_B T) / Z$ under the identification of phase amplitude with Boltzmann weight. The detailed correspondence is established in Section 19.

Together, these six formal results constitute the core mathematical content of the paper. They establish, in the order of logical dependence, that: the Phase Theory measure space is not itself a probability space (Prop. 29.1); probability spaces emerge by conditioning on preparations (Thm. 29.2); the Born rule follows from symplectic geometry (Thm. 29.3); no probabilistic primitive enters the derivation (Cor. 29.4); unpredictability is structural, not epistemic (Thm. 29.5); and the classical limit is a theorem, not an assumption (Thm. 29.6). These results form a logically tight structure that is self-consistent and derives all the key features of quantum and classical probability from the three axioms of Phase Theory.

30. Conclusion

The central thesis of this paper — that probability is not a primitive feature of physical law but a derived, emergent description of ensembles of admissible phase configurations under coarse-grained access and environmental coupling — has been established on a rigorous axiomatic and mathematical basis. Beginning with the three axioms of Phase Theory (Phase Ontology, Admissibility, and Global Consistency), and without invoking any probabilistic primitive, stochastic axiom, or collapse postulate, we have derived the complete probabilistic apparatus of quantum mechanics. The Born rule emerges as a theorem (Theorem 29.3) about the symplectic geometry of the admissible configuration space Conf_{adm} . The Kolmogorov probability axioms emerge (Theorem 29.2) as consequences of the measure-theoretic properties of the phase-geometric measure μ_{adm} . Unpredictability emerges (Theorem 29.5) as a structural consequence of the admissibility constraints on measurement interactions. The classical limit emerges (Theorem 29.6) as the regime of low topological complexity and strong environmental decoherence. Probability, in all its forms, is geometry.

The scope of the derivations presented here exceeds that of any previous attempt to derive probability within a physical theory. Gleason's theorem derives the Born rule from Hilbert space axioms, but these axioms already presuppose the quantum mechanical framework. Deutsch–Wallace decision-theoretic derivations derive the Born rule from rationality axioms, but these presuppose rational agents and a branching Everettian universe. Zurek's envariance argument derives the Born rule from symmetry arguments applied within quantum mechanics. Phase Theory's derivation is categorically more fundamental: it proceeds from axioms that do not presuppose quantum mechanics, and it derives not only the Born rule but also the Kolmogorov axioms themselves, the law of large numbers (as a geometric ergodicity theorem), the principle of decoherence, and the emergence of classical statistical mechanics. The derivation requires only the admissibility

functional, the natural symplectic measure, and the topological sector decomposition — all of which are intrinsic to Phase Theory and derivable from its axioms.

The experimental predictions of Phase Theory (Section 25) provide a program for empirical testing that distinguishes Phase Theory from standard quantum mechanics and from other quantum foundations programs. The five predictions — Born rule corrections in topological media, variance suppression in BECs, topology-dependent modification of the Tsirelson bound, non-Markovian memory effects in ensemble evolution, and gravitational correction to the decoherence rate — are all in principle testable with current or near-future experimental technology. Some (Born rule corrections in mesoscopic topological insulators, non-Markovian effects in long-coherence systems) are within the range of current experimental precision with appropriate experimental design. Phase Theory is not merely a philosophical reinterpretation of quantum mechanics; it is a new physical theory with novel empirical consequences.

The systematic comparison with existing quantum foundations programs (Section 28) has established Phase Theory's position in the landscape of interpretations and foundations proposals. Phase Theory is not an interpretation of quantum mechanics — it does not take quantum mechanics as given and ask what it means. It is a more fundamental theory from which quantum mechanics is derived as an effective description. It supersedes the measurement problem rather than solving it (the problem dissolves because its premises — the wave function as fundamental, the Born rule as primitive, collapse as a physical process — are absent from Phase Theory). It supersedes the probability debate among frequentists, Bayesians, and propensity theorists by providing the objective physical ground — the phase-geometric measure — that all three traditions have sought without being able to find within the probabilistic framework itself. Nature does not roll dice, as

Einstein famously hoped. But Phase Theory goes further: it explains why the dice appear to be rolled, and shows that the appearance is a theorem about geometry, not a fact about ontology.

The path forward is clear. Paper 6 of the Phase Theory series will provide the dynamical account of the measurement process, completing the demonstration that the full quantum mechanical formalism — state preparation, unitary evolution, measurement, Born rule probabilities, and post-measurement conditioning — is derivable from the three axioms of Phase Theory without additional postulates. The seven remaining papers of the series will extend the Phase Theory framework to the quantum theory of gravity, to the cosmological initial conditions, and to the experimental program required to test the novel predictions. The elimination of probability as a foundational concept in physics is not merely a philosophical achievement; it is the beginning of a new physical program that treats the geometry of the admissible configuration space as the fundamental substrate of all physical phenomena. Phase resolves; probability describes; geometry explains.

Appendix A: Measure Theory on Admissible Configuration Space

This appendix develops the rigorous measure-theoretic foundations for the phase-geometric measure μ_{adm} on the admissible configuration space Conf_{adm} . The constructions here are used throughout the main text and are presented with full mathematical detail.

Definition A.1 (Phase Topology and Borel σ -Algebra).

Let $\text{Conf}_{\text{adm}} = \{\Phi \in C^\infty(M, G) \mid I[\Phi] \geq 0\}$ be the admissible configuration space, equipped with the subspace topology inherited from the C^∞ topology on

$C^\infty(M, G)$ (uniform convergence of all derivatives on compact sets). The Borel σ -algebra $\mathcal{B}$ on Conf_adm is the σ -algebra generated by the open sets of this topology. A set $A \in \mathcal{B}$ is called a Borel set of Conf_adm .

Definition A.2 (Phase-Geometric Measure).

The phase-geometric measure μ_{adm} on $(\text{Conf_adm}, \mathcal{B})$ is defined as the Hausdorff measure of appropriate dimension on each topological sector $\text{Conf_adm}^{(n)}$, restricted to Conf_adm and extended to all of Conf_adm by σ -additivity over sectors. On each finite-dimensional approximation $\text{Conf_adm}^{(n, N)}$ (truncation to N Fourier modes in the sector with winding number n), μ_{adm} is defined as the Liouville measure $\omega^N/N!$ where ω is the symplectic form defined in equation (8.1). The full measure is the projective limit as $N \rightarrow \infty$, subject to the regularity conditions of Lemma A.1.

Lemma A.1 (σ -Finiteness of μ_{adm}).

The phase-geometric measure μ_{adm} is σ -finite on $(\text{Conf_adm}, \mathcal{B})$ if and only if the admissibility density $L(\Phi, \nabla\Phi, \kappa)$ satisfies: (i) L is lower semi-continuous in the C^∞ topology; (ii) The sublevel sets $\{\Phi \mid I[\Phi] \leq C\}$ are compact in the C^∞ topology for every $C > 0$; and (iii) The winding number functional $n[\Phi]$ is continuous on Conf_adm . Under these conditions, $\text{Conf_adm} = \sqcup_n \text{Conf_adm}^{(n)}$ is a countable disjoint union of σ -finite measure spaces, making $(\text{Conf_adm}, \mathcal{B}, \mu_{\text{adm}})$ σ -finite.

Proof sketch. Condition (i) ensures that Conf_adm is a closed subset of $C^\infty(M, G)$ and that $I[\Phi]$ is measurable. Condition (ii) ensures that each sublevel set has finite μ_{adm} -measure, giving σ -finiteness within each sector. Condition (iii) ensures that the sector decomposition is measurable. The countability of the sector index set ($n \in \mathbb{Z}$) gives the σ -finiteness of the full measure space. ■

Lemma A.2 (Disintegration Theorem for μ_{adm}).

Under the conditions of Lemma A.1, the phase-geometric measure μ_{adm} disintegrates over the topological sector decomposition $\text{Conf}_{\text{adm}} = \sqcup_n \text{Conf}_{\text{adm}}^{(n)}$ as: $\mu_{\text{adm}} = \sum_n \mu_{\text{adm}}^{(n)}$, where $\mu_{\text{adm}}^{(n)} = \mu_{\text{adm}}|_{\{\text{Conf}_{\text{adm}}^{(n)}\}}$ is the restriction of μ_{adm} to the n -th sector. Each $\mu_{\text{adm}}^{(n)}$ is a positive, σ -finite measure on the sector $\text{Conf}_{\text{adm}}^{(n)}$ with its induced Borel structure. For any preparation P , the disintegration restricts to: $\mu_{\text{adm}}|_{\{E(P)\}} = \sum_n \mu_{\text{adm}}^{(n)}|_{\{E_n(P)\}}$, where $E_n(P) = E(P) \cap \text{Conf}_{\text{adm}}^{(n)}$ is the n -th sector component of the admissible ensemble.

The disintegration theorem of Lemma A.2 is the measure-theoretic foundation for the Born rule derivation: it justifies the decomposition of the admissible ensemble into sectors with individually defined phase-geometric measures, which are then the inputs to the sector-volume computation in Appendix B. The conditions of Lemma A.1 are satisfied by the admissibility functional $I[\Phi]$ defined in equation (4.2) under mild regularity assumptions on the admissibility density L , which are verified for all physically relevant cases in Paper 1 of the Phase Theory series.

Appendix B: Born Rule Derivation — Detailed Computation

This appendix presents the complete step-by-step derivation of the Born rule from the symplectic structure of the admissible configuration space, filling in the details of the proof sketch given in Section 9.

Step 1: The symplectic form on Conf_{adm} . The symplectic form ω on Conf_{adm} is defined by equation (8.1): $\omega(\delta_1\Phi, \delta_2\Phi) = \int_M \text{Im}(\delta_1\Phi^* \cdot \delta_2\Phi) d\mu_M$. For a sector decomposition $\Phi = \sum_k A_k \Phi_k$, the perturbations decompose as $\delta\Phi = \sum_k \delta A_k \Phi_k$ (within the finite-dimensional amplitude

space) plus transverse perturbations within each sector. The transverse perturbations within each sector are orthogonal to the amplitude directions in the symplectic form (by the sector decomposition being orthogonal in the symplectic sense: $\omega(\Phi_j, \Phi_k) = \delta_{jk}$), so the symplectic form on the amplitude space is:

$$\omega|_{\text{amp}} = (i/2) \sum_k dA_k \wedge d\bar{A}_k = (1/2) \sum_k d|A_k|^2 \wedge d\theta_k \quad (\text{B.1})$$

where $\theta_k = \arg(A_k)$. This is the standard Kähler form on $\mathbb{C}^K$ (for K sectors), restricted to the unit sphere $\sum_k |A_k|^2 = 1$ (normalization of phase amplitudes).

Step 2: Sector volume computation. The symplectic volume of the k -th sector component E_k of the admissible ensemble is the integral of the Liouville measure $\omega^{K-1}/(K-1)!$ over E_k , restricted to the unit sphere $\sum_k |A_k|^2 = 1$. This is a standard computation on complex projective space $\mathbb{CP}^{K-1}$. The unit sphere in $\mathbb{C}^K$, equipped with the Fubini-Study metric derived from $\omega|_{\text{amp}}$, decomposes into fibers parameterized by the overall phase $\sum_k \theta_k$ (which is redundant due to the $U(1)$ action on the amplitude space). The invariant measure on the sphere, after quotienting by this $U(1)$, is the Fubini-Study measure $d\text{FS} = (K-1)! \prod_{k=1}^K d|A_k|^2 \cdot \delta(\sum_k |A_k|^2 - 1)$. The volume of the k -th sector (the subset of the sphere where the k -th amplitude is the "dominant" one, or more precisely the measure of the k -th sector in the full sphere) is:

$$\text{Vol}(E_k) = \int_{E_k} d\text{FS} = |A_k|^2 \cdot \text{Vol}(S^{2K-1}) \quad (\text{B.2})$$

This equality holds because the Fubini-Study measure on $\mathbb{CP}^{K-1}$ assigns measure $|A_k|^2$ to the " k -th coordinate patch," as a consequence of the $U(K)$ invariance of the Fubini-Study metric and the fact that the k -th coordinate direction has squared norm $|A_k|^2$. The computation is identical to the calculation of the area of a spherical zone in S^{2K-1} .

Step 3: Normalization and the Born Rule. Dividing equation (B.2) by the total volume $\text{Vol}(E(P)) = \sum_k \text{Vol}(E_k) = \sum_k |A_k|^2 \cdot \text{Vol}(S^{\{2K-1\}}) = \text{Vol}(S^{\{2K-1\}})$ (since $\sum_k |A_k|^2 = 1$ by normalization), the Born Rule probability follows:

$$P(k | P) = \text{Vol}(E_k) / \text{Vol}(E(P)) = |A_k|^2 \cdot \text{Vol}(S^{\{2K-1\}}) / \text{Vol}(S^{\{2K-1\}}) = |A_k|^2 \quad (\text{B.3})$$

Step 4: Continuous spectrum extension. For a continuous observable with spectrum $\{k \in \mathbb{R}\}$, the sector decomposition becomes a continuous family of sectors parameterized by k , and the phase amplitudes A_k become an amplitude density $A(k) \in \mathbb{C}$ with $\int |A(k)|^2 dk = 1$. The Born Rule probability for $k \in [k_0, k_1]$ is $P(k \in [k_0, k_1] | P) = \int_{k_0}^{k_1} |A(k)|^2 dk$, which is the standard Born Rule for continuous observables. The derivation follows by replacing sums with integrals in equations (B.1)–(B.3), using the Lebesgue integral on the continuous amplitude space.

Step 5: Interference term from sector boundary overlap. When sectors E_j and E_k have overlapping boundaries $B_{\{jk\}}$ with $\mu_{\text{adm}}(B_{\{jk\}}) > 0$, the projection π cannot cleanly separate the E_j and E_k contributions to an outcome in the overlap region. The symplectic volume computation for outcomes in the boundary region yields a contribution from the cross-term $\omega(A_j \Phi_j, A_k \Phi_k) = \text{Re}(A_j^* A_k) \omega(\Phi_j, \Phi_k)$. For orthogonal sectors ($\omega(\Phi_j, \Phi_k) = 0$), there is no cross-term and no interference. For sectors with boundary overlap ($\omega(\Phi_j, \Phi_k) \neq 0$ in the boundary region), the cross-term contributes to the probability of outcomes in the overlap region as $2 \text{Re}(A_j^* A_k)$, which is the quantum mechanical interference term. Constructive interference corresponds to $\text{Re}(A_j^* A_k) > 0$; destructive interference to $\text{Re}(A_j^* A_k) < 0$. This completes the derivation of the full Born rule, including interference.

Appendix C: Glossary of Phase Theory Terms

This appendix defines the principal technical terms of Phase Theory as used throughout this paper. Definitions are given in full prose, following the formal conventions established in Sections 2 through 9.

The ***phase field*** Φ is the sole primitive ontological constituent of physical reality in Phase Theory. It is a smooth map $\Phi : M \rightarrow G$ from the topological substrate M (a compact, connected, orientable manifold without boundary) to a compact Lie group G (typically $U(1)$ for electrodynamics, or $U(1) \times SU(2) \times SU(3)$ for the full standard model). All physical structures — particles, forces, spacetime geometry, and quantum states — are emergent features of the global behavior of Φ .

The ***admissibility functional*** $I[\Phi]$ is the real-valued functional on the space of phase configurations $C^\infty(M, G)$ defined by $I[\Phi] = \int_M L(\Phi, \nabla\Phi, \kappa) d\mu_M$, where L is the admissibility density incorporating the phase coherence length κ , local self-interaction $V(\Phi)$, and topological weighting $W(n[\Phi])$. It plays the role of a filter on the space of all configurations: configurations with $I[\Phi] \geq 0$ are admissible; those with $I[\Phi] < 0$ are not physically realizable. It is not a Hamiltonian or action functional.

An ***admissible configuration*** is any element Φ of the function space $C^\infty(M, G)$ satisfying $I[\Phi] \geq 0$. Admissible configurations constitute the space of all physically realizable states of the phase field. Each admissible configuration is fully determinate — it specifies the phase field at every point of M — and is not a superposition, probability amplitude, or wave function.

A ***phase sector*** is a connected component $\text{Conf_adm}^\wedge(n)$ of the admissible configuration space Conf_adm , indexed by the topological winding number $n[\Phi] \in \mathbb{Z}$. Phase sectors are the geometric precursors of particle species and measurement outcome categories in Phase Theory. Configurations within the same sector can be continuously deformed into each other (through admissible configurations) without changing the winding number. Configurations in different sectors are topologically distinct and cannot be continuously related within Conf_adm .

The ***topological winding number*** $n[\Phi]$ is the integer-valued topological invariant of the phase configuration Φ , defined by the degree of the map $\Phi : M \rightarrow G$. For $G = U(1)$, it is $n[\Phi] = (1/2\pi) \oint_C d\phi$ for an appropriate closed curve C in M . It is

conserved under continuous deformations of Φ within Conf_adm and takes only integer values, providing the topological origin of the discreteness of physical quantities.

The ***admissible ensemble*** $E(P)$ associated with a preparation procedure P is the set of all admissible configurations consistent with the constraints specified by P : $E(P) = \{\Phi \in \text{Conf_adm} \mid \Phi \text{ satisfies all constraints } C_\alpha \text{ of } P\}$. It is the Phase Theory analog of the Gibbs ensemble in statistical mechanics, but defined purely by admissibility and preparation constraints, without any probabilistic postulate. Each element $\Phi_i \in E(P)$ is a fully determinate physical state; the ensemble captures the underdetermination of the full phase configuration by the preparation procedure.

The ***phase-geometric measure*** μ_{adm} is the natural geometric measure on Conf_adm , derived from the symplectic form ω defined by equation (8.1). On each finite-dimensional approximation (truncation to N Fourier modes), it is the Liouville measure $\omega^N/N!$. In the full infinite-dimensional setting, it is the projective limit of these measures, subject to the regularity conditions of Lemma A.1. It is not a probability measure (it is unnormalized on all of Conf_adm) but becomes a probability measure upon restriction to an admissible ensemble $E(P)$ and normalization by $\mu_{\text{adm}}(E(P))$.

The ***coarse-graining projection*** π is a measurable map $\pi : \text{Conf_adm} \rightarrow O$ from the admissible configuration space to the finite or countable set of macroscopically distinguishable observable outcomes O . It is determined by the physical coupling between the measuring apparatus A and the system S : $\pi(\Phi_i) = O_j$ if the admissibility-constrained evolution of the composite $\Phi_{\{i,S+A\}}$ through the measurement interaction produces the apparatus record O_j . The projection π is many-to-one (multiple admissible configurations produce the same outcome) and context-dependent (different apparatus configurations produce different projections).

Environmental coupling refers to the admissibility-constrained interaction between the system S and the environment E — the totality of phase degrees of freedom in Conf_adm not accessible to the measuring apparatus. Environmental coupling entangles (in the phase-theoretic sense: correlates the topological sector assignments of) the system configuration with the environment, reducing the

effective accessible sector of Conf_adm . Strong environmental coupling produces decoherence: the suppression of sector cross-terms in the reduced phase-geometric measure on E_S , and the emergence of effectively classical probability distributions over definite outcomes.

Sector overlap is the phenomenon in which the boundaries of two distinct phase sectors E_j and E_k share a common sub-manifold $B_{\{jk\}} = \partial E_j \cap \partial E_k$ of positive phase-geometric measure. Sector overlap is the Phase Theory account of quantum superposition and interference: when sectors overlap, the coarse-graining projection π cannot cleanly separate their contributions to observable outcomes, producing cross-terms in the probability distribution that correspond to quantum mechanical interference terms. Constructive interference arises from phase-aligned overlap ($\text{Re}(A_j^* A_k) > 0$ at $B_{\{jk\}}$); destructive interference from anti-aligned overlap ($\text{Re}(A_j^* A_k) < 0$ at $B_{\{jk\}}$).

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